

Quiz:



= 17



= 18



= 20



= ?

- 0: background
- 1: person
- 2: bicycle
- 3: car
- 4: motorcycle
- 5: airplane
- 6: bus
- 7: train
- 8: truck
- 9: boat
- 10: traffic light
- 11: fire hydrant
- 12: stop sign
- 13: parking meter
- 14: bench
- 15: bird
- 16: cat
- 17: dog
- 18: horse
- 19: sheep
- 20: cow
- 21: elephant
- 22: bear
- 23: zebra
- 24: giraffe

MS COCO Taxonomy

Vision and Language Models

September 25th, 2025

Joungbin An

Era of Vision and Language



Heuristics:

ID: 16 → cat

Associate image with generic text:

Cat

Cute kitten

Curious kitty

Soft fur

Adorable pet

An image of a cat

ChatGPT 5 ▾



How would you describe this cat? What can you infer from this image?

This image shows a **small, orange (ginger) kitten** with white markings on its chest and paws. It has **large, bright eyes** and a curious, slightly cautious expression. The kitten is sitting upright on a stone surface with **green grass in the background**, suggesting the photo was taken outdoors or in a garden-like setting.

From the image, one could infer:

- The kitten is very **young**, likely just a few weeks to a couple of months old, judging by its size and facial features.
- Its upright posture and direct gaze show it may be **alert and attentive**, possibly noticing the photographer.
- The setting looks **calm and natural**, which enhances the impression of innocence and playfulness.
- Its clean fur and healthy appearance suggest it is **well cared for**.

Would you like me to generate different *tones of descriptions*—like one that's factual, one that's cute/affectionate, and one that's a bit more poetic?

LXMERT: Learning Cross-Modality Encoder Representations from Transformers

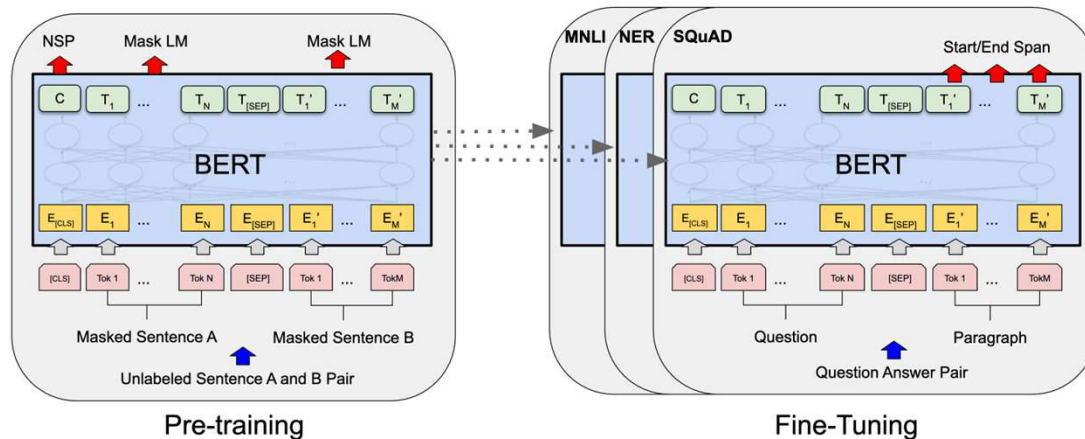
Hao Tan Mohit Bansal

UNC Chapel Hill

EMNLP 2019

LXMERT: Background

By 2019, BERT-style pretraining had transformed NLP



But vision-language systems were still largely **task-specific** and **data-hungry**

- No general, pretrained VL backbone analogous to BERT
- Weak transfer across tasks
- Poor cross-modal alignment between words and visual entities

LXMERT: Objective

LXMERT's Question:

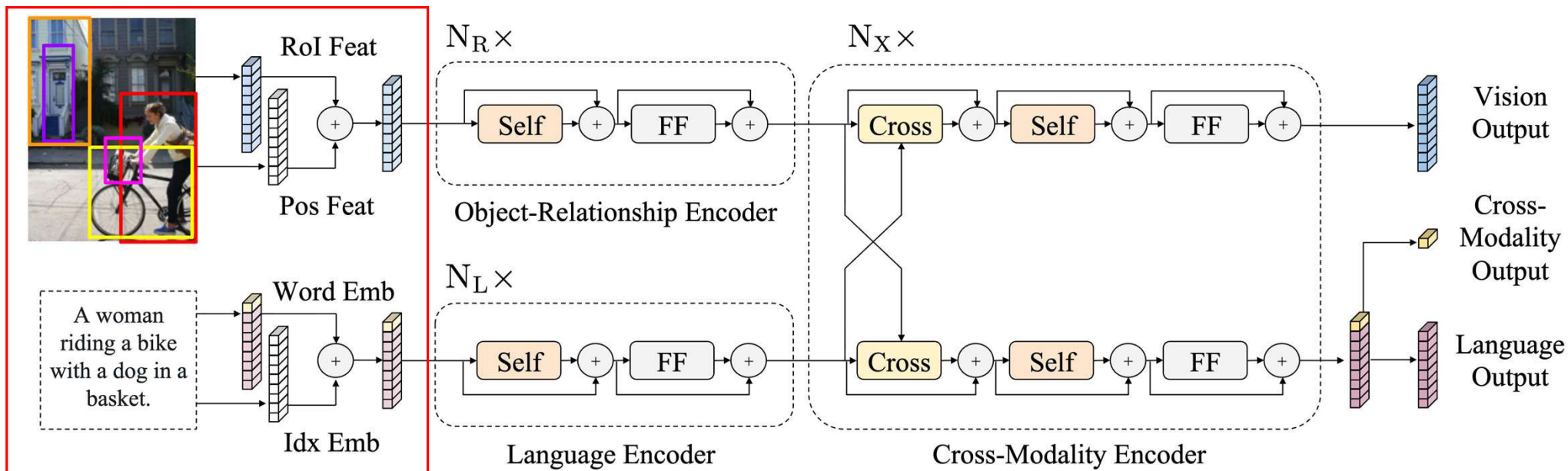
Can you build a **single, reusable Vision Language backbone** with strong performance on **diverse** VL tasks?

Learn **general-purpose, cross-modal representations** by **pretraining a Transformer** that:

- a) Understands language
- b) Understands object-level visual scenes
- c) Aligns the two through cross-attention

Pretrain once, align language ↔ vision, transfer everywhere!

LXMERT: Method



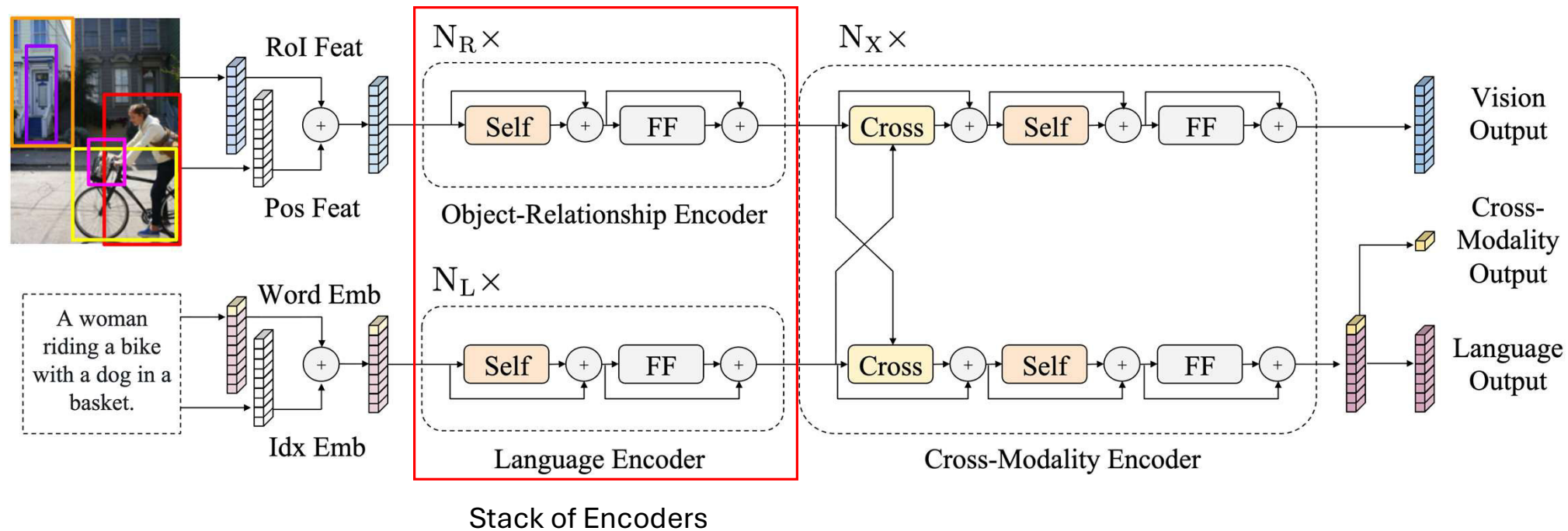
Object Features through Faster-RCNN¹

Word embeddings produced following BERT²

[1] Ren, Shaoqing, et al. "Faster r-cnn: Towards real-time object detection with region proposal networks." NeurIPS 2015.

[2] Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." NAACL 2019.

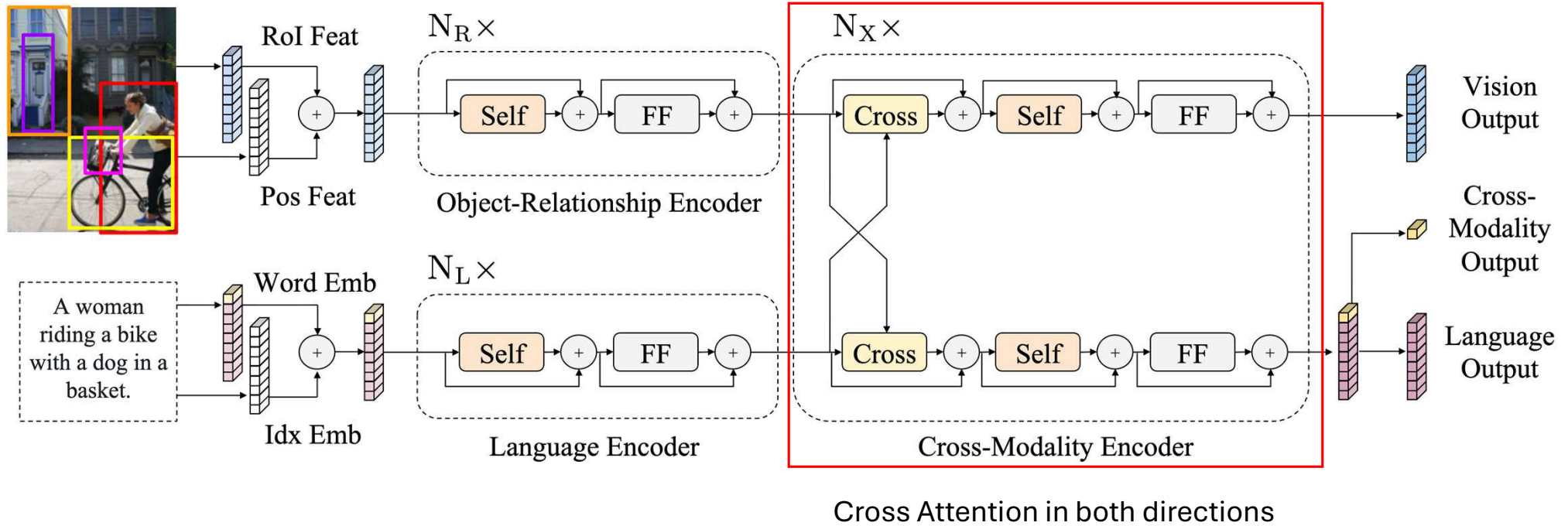
LXMERT: Method



[1] Ren, Shaoqing, et al. "Faster r-cnn: Towards real-time object detection with region proposal networks." NeurIPS 2015.

[2] Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." NAACL 2019.

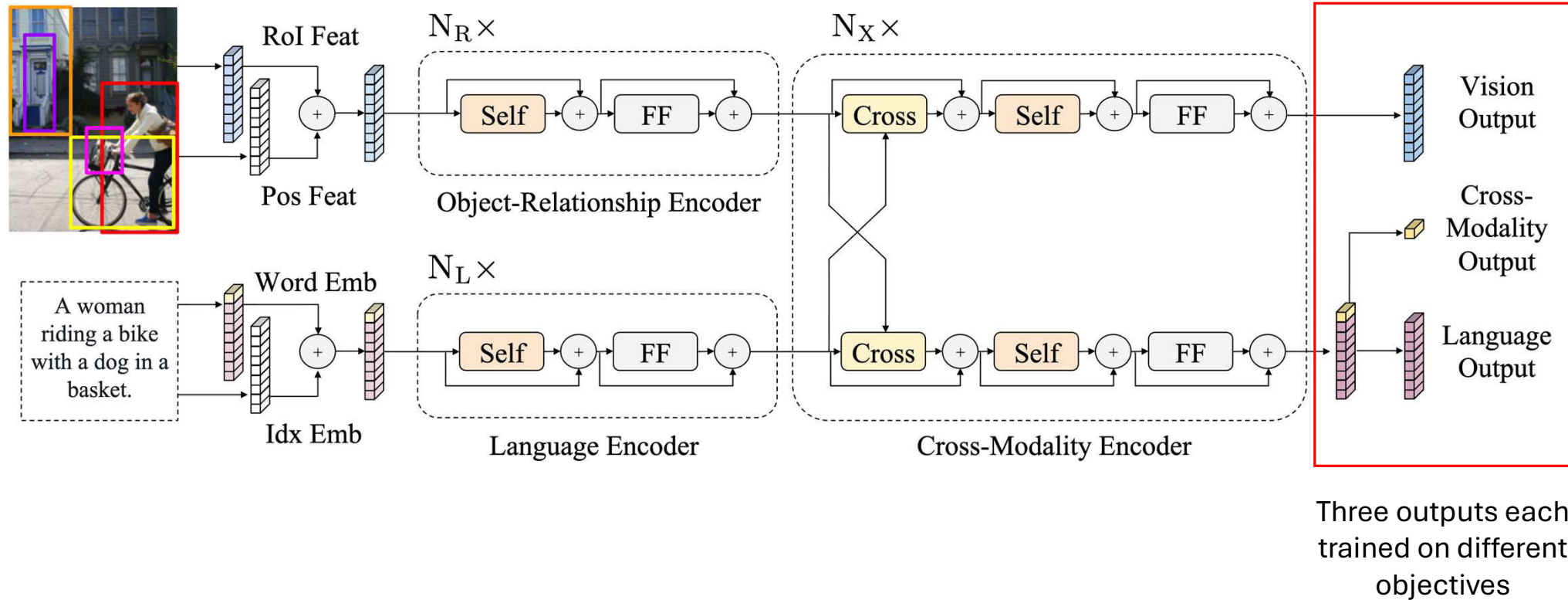
LXMERT: Method



[1] Ren, Shaoqing, et al. "Faster r-cnn: Towards real-time object detection with region proposal networks." NeurIPS 2015.

[2] Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." NAACL 2019.

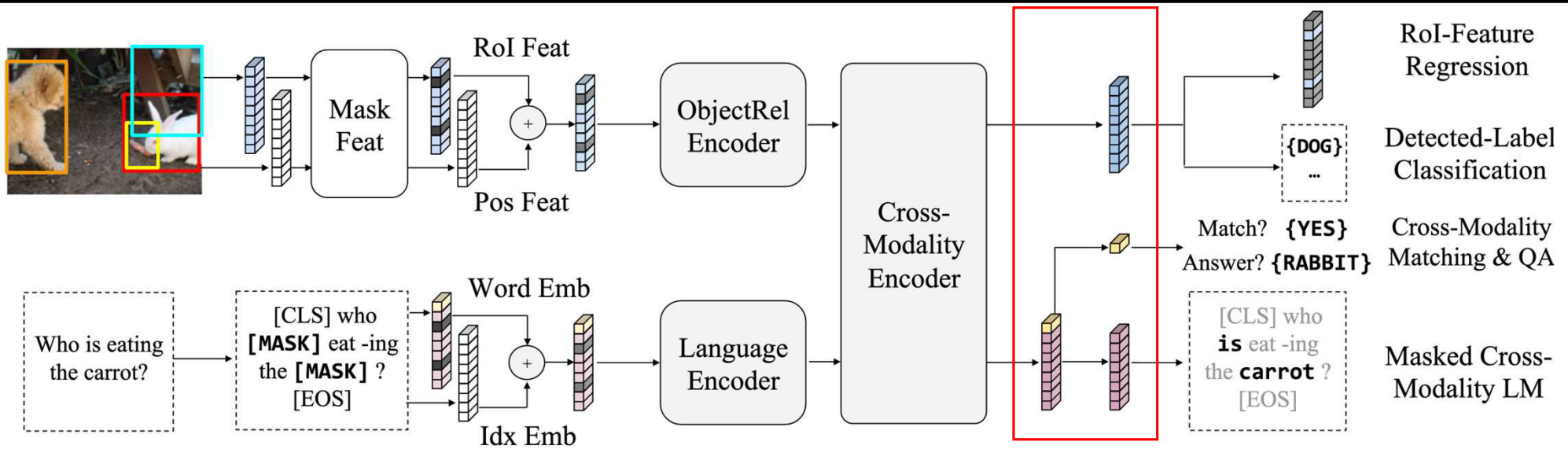
LXMERT: Method



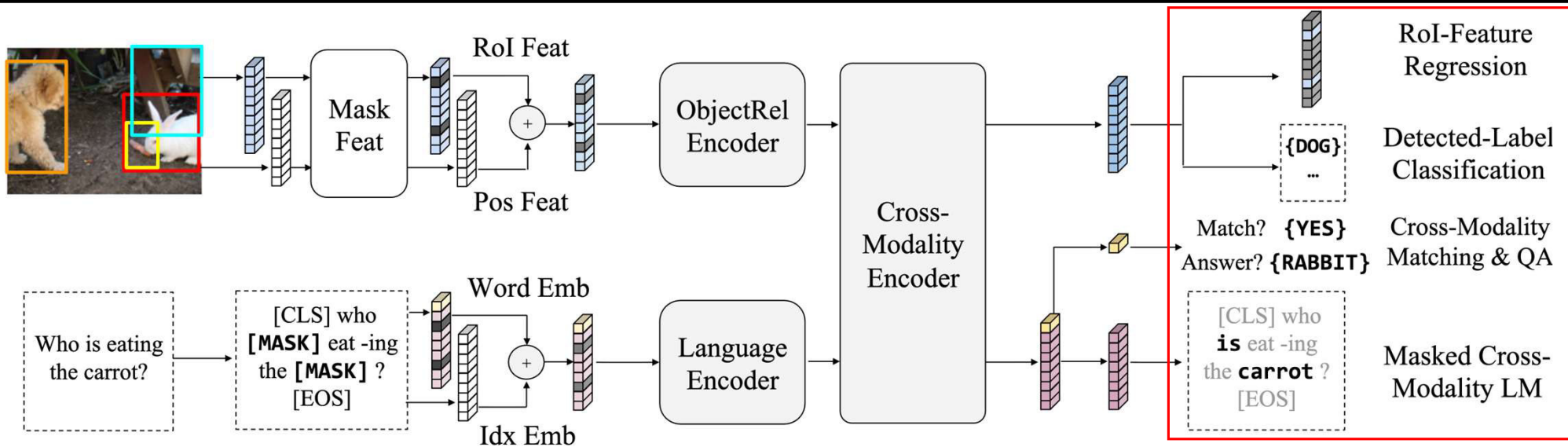
Three outputs each
trained on different
objectives

[1] Ren, Shaoqing, et al. "Faster r-cnn: Towards real-time object detection with region proposal networks." NeurIPS 2015.
[2] Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." NAACL 2019.

LXMERT: Method - Pretraining



LXMERT: Method - Pretraining



Multiple Pretraining tasks from the 5 aggregated vision-language datasets

Datasets used:

Image Split	Images	Sentences (or Questions)					
		COCO-Cap	VG-Cap	VQA	GQA	VG-QA	All
MS COCO - VG	72K	361K	-	387K	-	-	0.75M
MS COCO $\cap$ VG	51K	256K	2.54M	271K	515K	724K	4.30M
VG - MS COCO	57K	-	2.85M	-	556K	718K	4.13M
All	180K	617K	5.39M	658K	1.07M	1.44M	9.18M

LXMERT: Results

Single model to solve ALL!

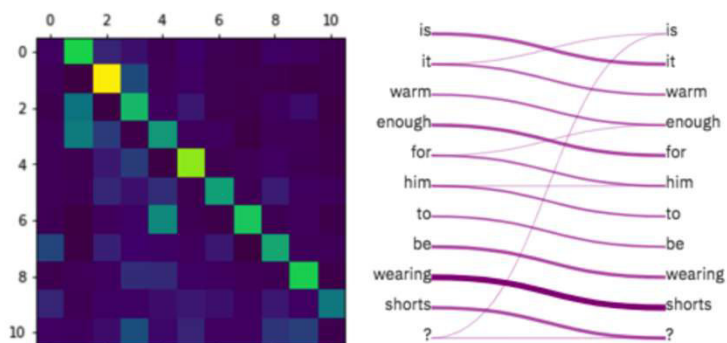
Method	VQA				GQA			NLVR ²	
	Binary	Number	Other	Accu	Binary	Open	Accu	Cons	Accu
Human	-	-	-	-	91.2	87.4	89.3	-	96.3
Image Only	-	-	-	-	36.1	1.74	17.8	7.40	51.9
Language Only	66.8	31.8	27.6	44.3	61.9	22.7	41.1	4.20	51.1
State-of-the-Art	85.8	53.7	60.7	70.4	76.0	40.4	57.1	12.0	53.5
LXMERT	88.2	54.2	63.1	72.5	77.8	45.0	60.3	42.1	76.2

Method	VQA	GQA	NLVR ²
1. No Vision Tasks	66.3	57.1	50.9
2. Feat	69.2	59.5	72.9
3. Label	69.5	59.3	73.5
4. Feat + Label	69.9	60.0	74.9

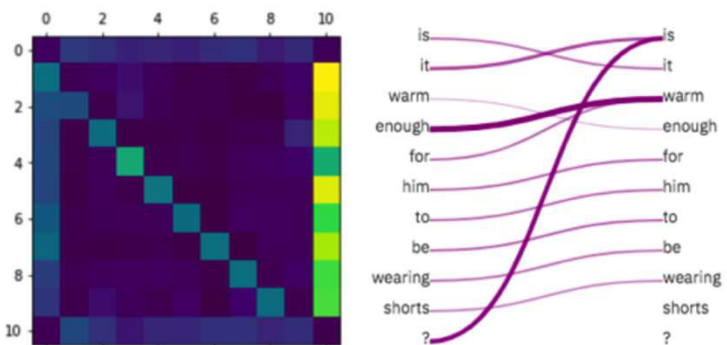
Vision pretraining tasks are important!

Table 5: Dev-set accuracy of different vision pre-training tasks. ‘Feat’ is RoI-feature regression; ‘Label’ is detected-label classification.

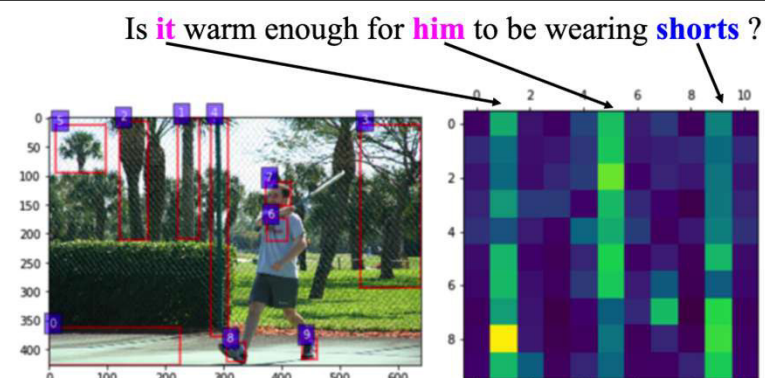
LXMERT: Visualizations



(a) LXMERT 2nd Lang-layer (b) BERT 3rd Layer



(c) LXMERT 4th Lang-layer (d) BERT 4th Layer



What colors are **the pole** **the horse** is jumping over?

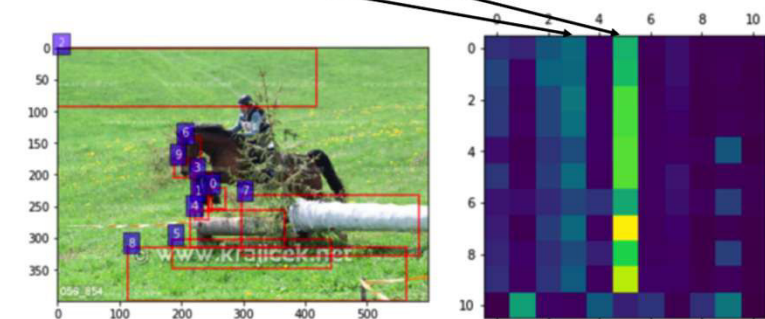


Figure 5: Attention graphs in LXMERT's cross-modality encoder showing that the attention focus on pronouns (marked in pink), nouns (marked in blue), and articles (marked in red).

LXMERT: Summary

Shows A **single, reusable VL backbone** with strong performance on diverse VL tasks can be unlocked through adequate vision-language pretraining

Limitations:

- **Detector-dependent region features (not end-to-end pixels)**
 - Susceptible to detector biases/errors
 - Noisy supervision for object labels
- **Closed answer space for QA**
 - Pretraining uses a 9.5k-answer table (~90% coverage), limiting open-ended generation.

Learning Transferable Visual Models From Natural Language Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

ICML 2021

CLIP: Background

Scaling laws in NLP showed that pretraining on broad, weakly supervised text yields transferable language models.

Conventional vision pretraining (e.g., ImageNet classification) uses fixed, human-curated label sets. That creates three bottlenecks:

1. **Narrow supervision**: single labels per image don't capture rich semantics (attributes, relations, long-tail concepts).
2. **Poor transfer**: models overfit to a dataset's taxonomy and degrade under distribution shift and new label spaces.
3. **Scaling friction**: expanding to more tasks requires new annotation.

CLIP: Objective

CLIP's Question:

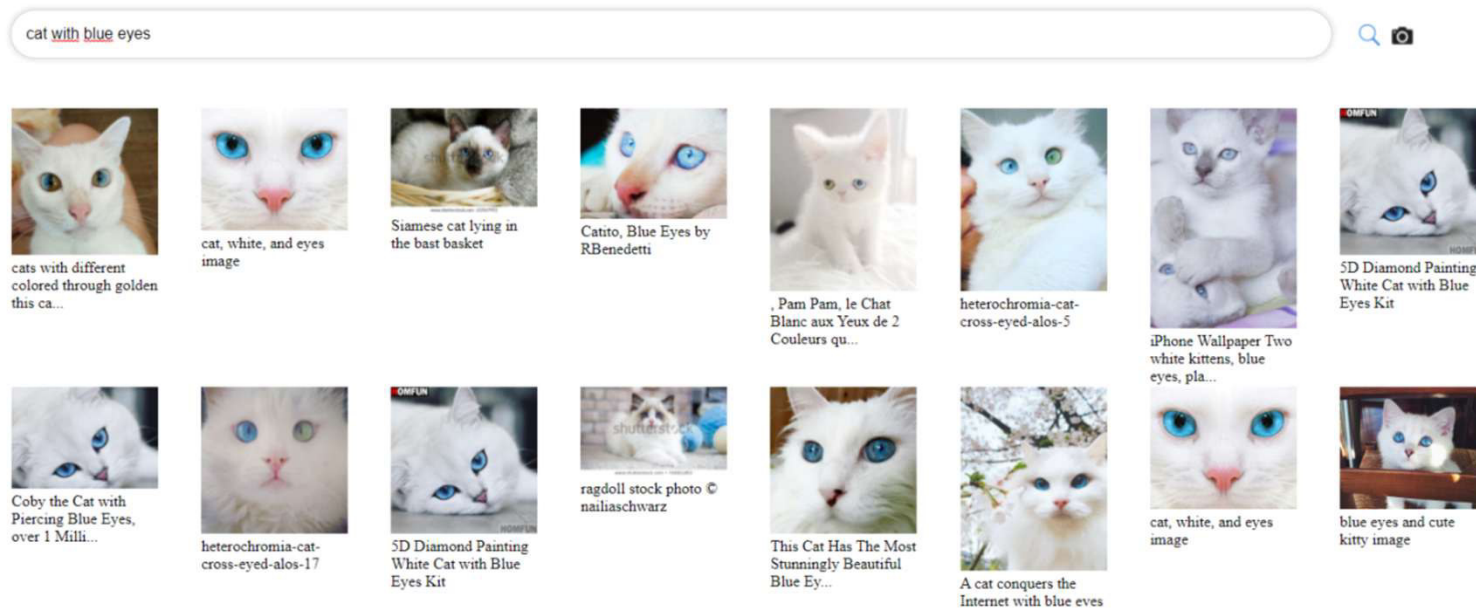
*Can we replace **fixed labels** with the **open-ended supervision** already embedded in natural language, so that a **single pretrained model zero-shots** to many tasks simply by changing the text prompt?*

Learn a universal vision–language model that can recognize anything from just a prompt!

CLIP: Method

Dataset: Collect 400 million image-text pairs scraped from the public internet

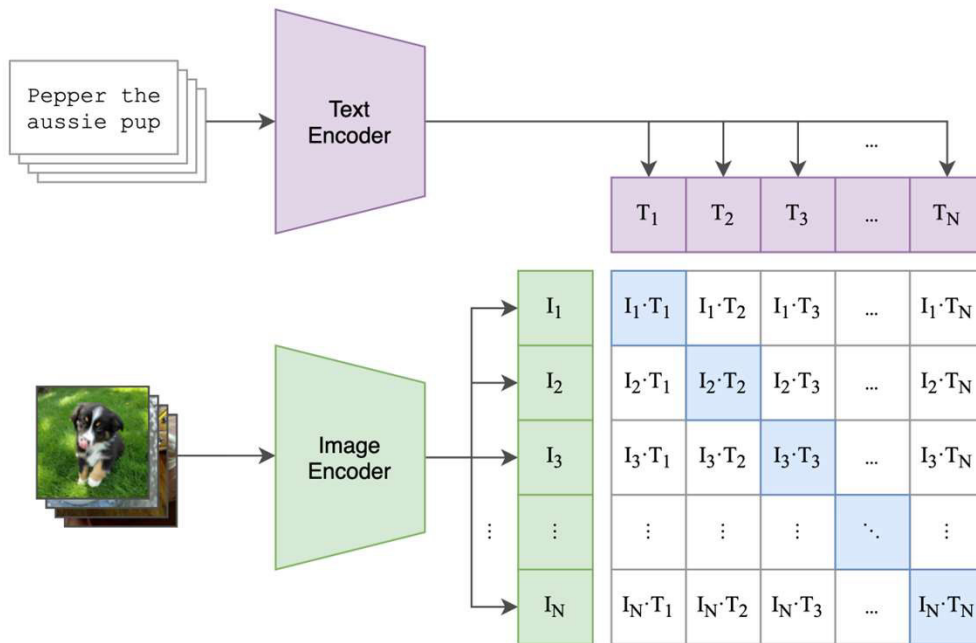
- Ensure text is not too short and images not too small, but keep dataset noisy to maintain scale



CLIP: Method

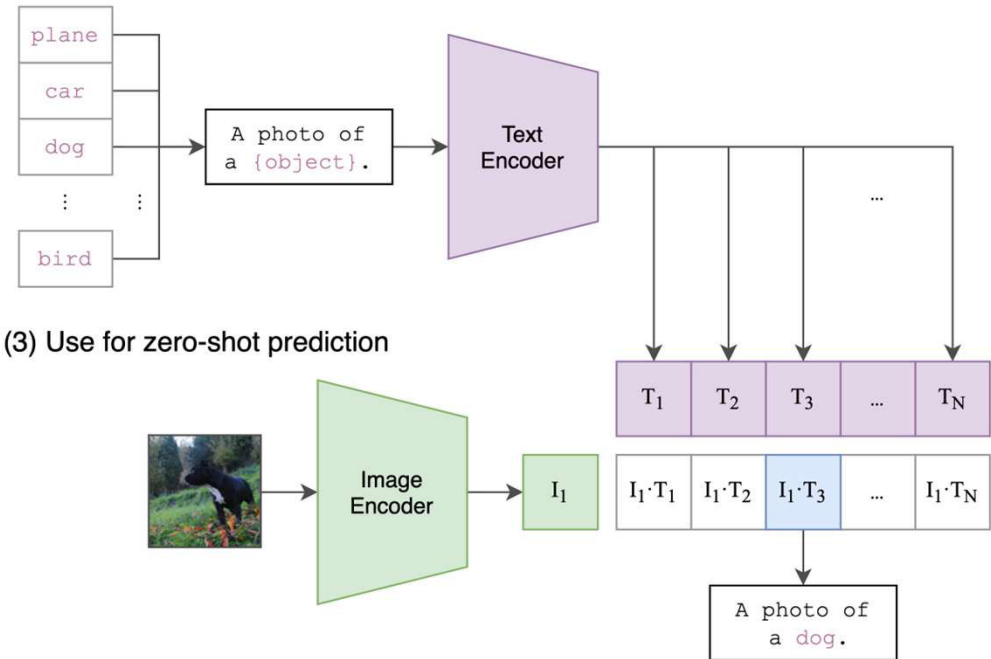
Contrastive Pre-training on collected data

(1) Contrastive pre-training

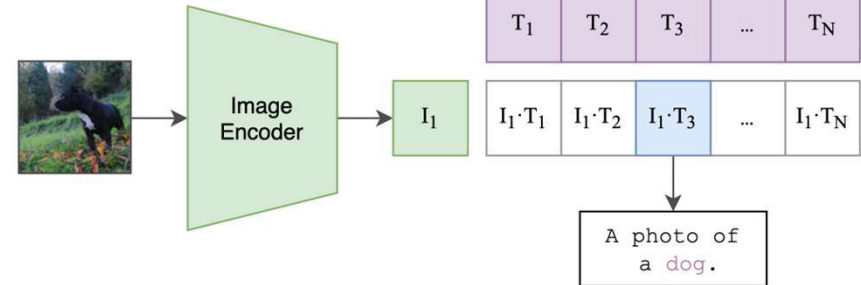


Maximize the similarity with the matched pairs

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



Argmax cosine similarity for prediction

CLIP: Results

Outperforms many fully supervised and outperforms few-shot trained models!

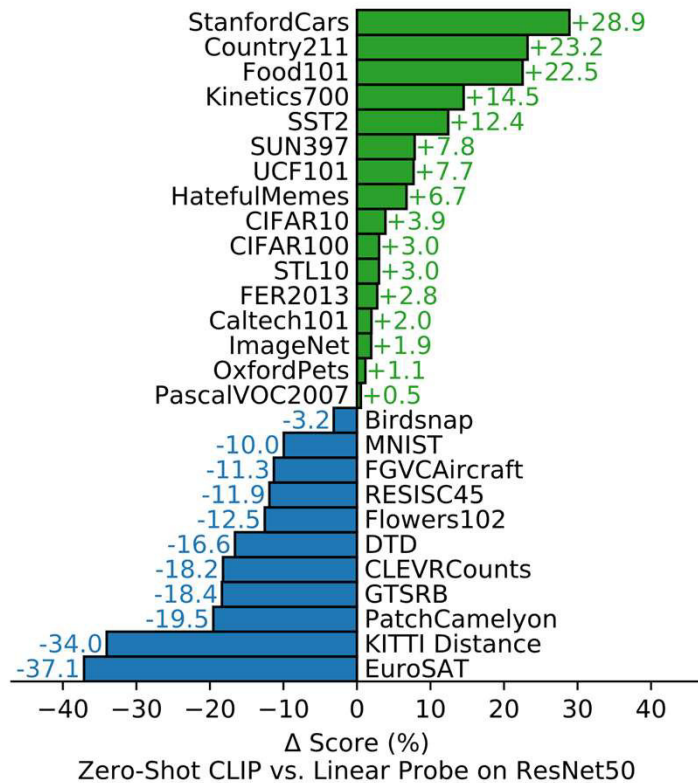


Figure 5. Zero-shot CLIP is competitive with a fully supervised baseline. Across a 27 dataset eval suite, a zero-shot CLIP classifier outperforms a fully supervised linear classifier fitted on ResNet-50 features on 16 datasets, including ImageNet.

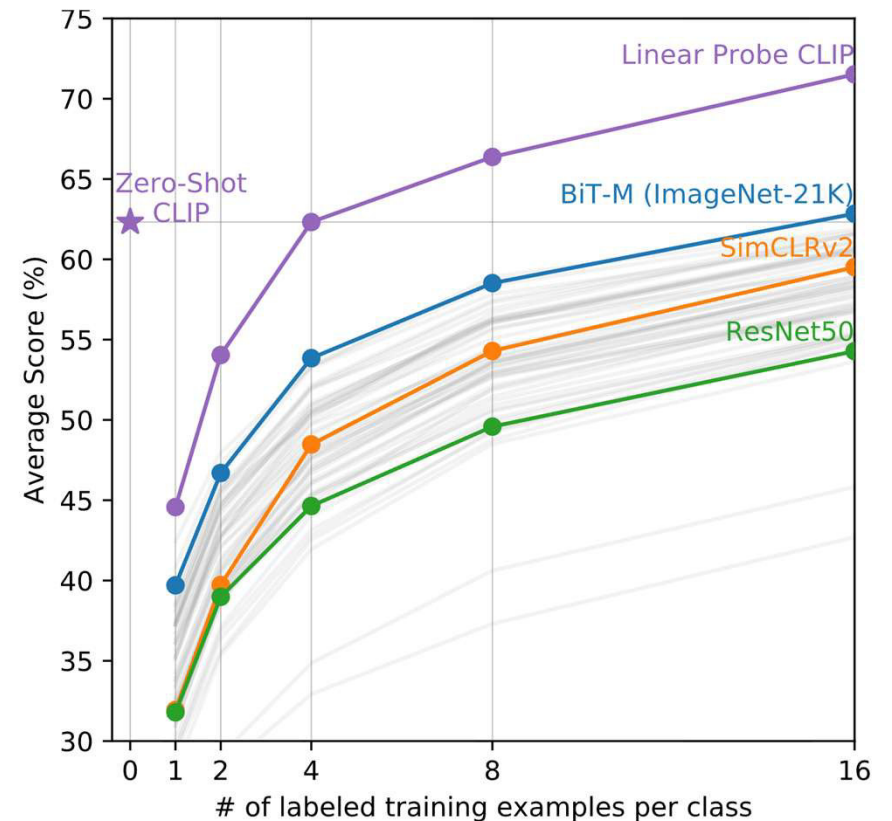
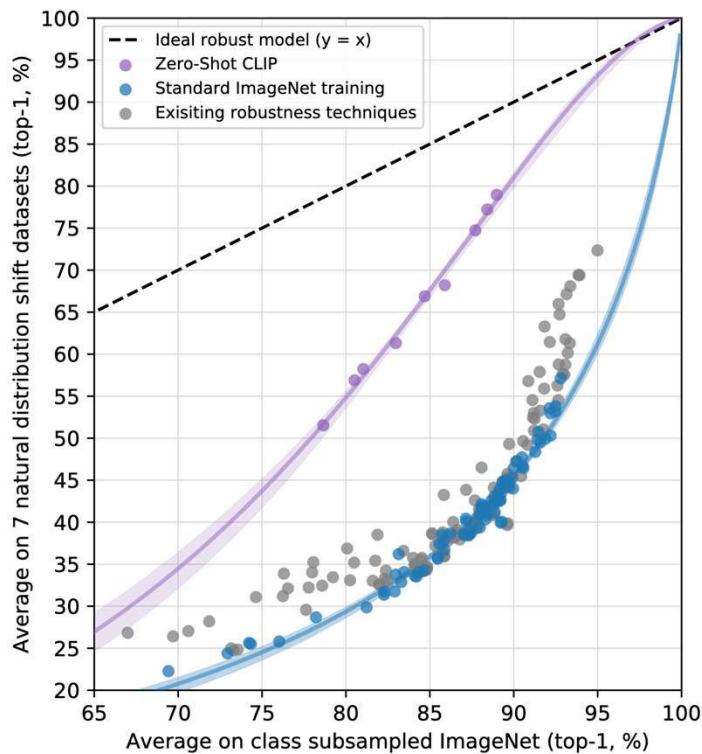


Figure 6. Zero-shot CLIP outperforms few-shot linear probes.

CLIP: Results

Zero-shot CLIP is much more **robust to distribution shift** than standard ImageNet models



	Dataset Examples						ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet							76.2	76.2	0%
ImageNetV2							64.3	70.1	+5.8%
ImageNet-R							37.7	88.9	+51.2%
ObjectNet							32.6	72.3	+39.7%
ImageNet Sketch							25.2	60.2	+35.0%
ImageNet-A							2.7	77.1	+74.4%

CLIP Visualization

ImageNet

King Charles Spaniel (91.6%) Ranked 1 out of 1000 labels



✓ a photo of a **king charles spaniel**.

✗ a photo of a **brittany dog**.

✗ a photo of a **cocker spaniel**.

✗ a photo of a **papillon**.

✗ a photo of a **sussex spaniel**.

aYahoo

building (97.7%) Ranked 1 out of 12 labels



✓ a photo of a **building**.

✗ a photo of a **carriage**.

✗ a photo of a **statue**.

✗ a photo of a **bag**.

✗ a photo of a **mug**.

Birdsnap

Black chinned Hummingbird (33.3%) Ranked 4 out of 500 labels



✗ a photo of a **broad tailed hummingbird**, a type of bird.

✗ a photo of a **calliope hummingbird**, a type of bird.

✗ a photo of a **costas hummingbird**, a type of bird.

✓ a photo of a **black chinned hummingbird**, a type of bird.

✗ a photo of a **annas hummingbird**, a type of bird.

ObjectNet ImageNet Overlap

Pill bottle (98.3%) Ranked 1 out of 113 labels



✓ a photo of a **pill bottle**.

✗ a photo of a **bottle cap**.

✗ a photo of a **beer bottle**.

✗ a photo of a **pillow**.

✗ a photo of a **wine bottle**.

CLIP: Summary

Open-vocabulary zero-shot VL model with strong results across tasks/datasets that is **robust** under distribution shift.

- A LOT of engineering (gradient check pointing, precision, etc). Please check the full paper.

Limitations:

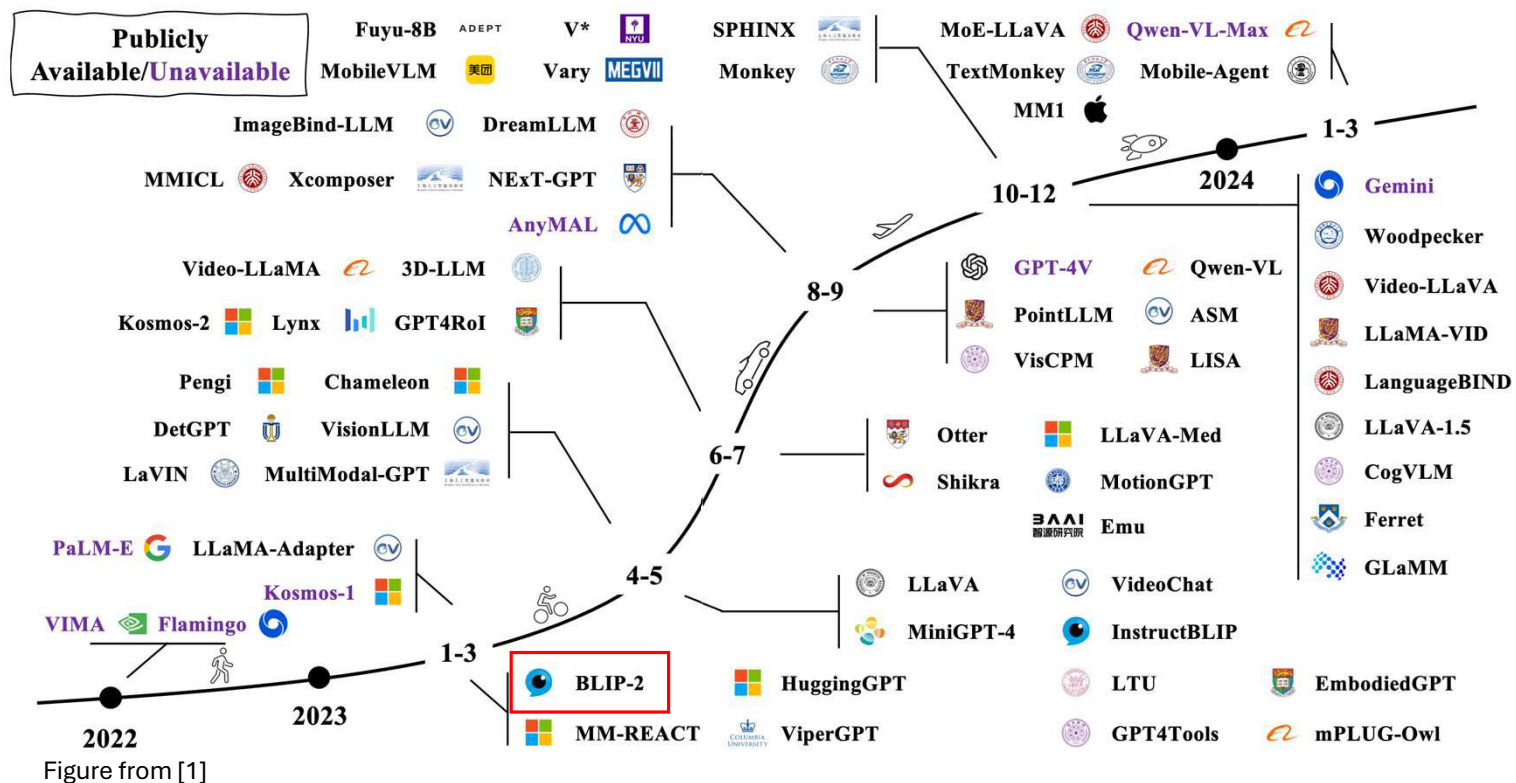
- Prompt Dependent
 - Relies on hand-crafted class prompts/ensembles
- Web data bias
 - Training pairs are unfiltered/uncurated and CLIP inherits social biases
- Non-generative
 - Cannot generate language or follow arbitrary instructions

BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models

Junnan Li Dongxu Li Silvio Savarese Steven Hoi
Salesforce Research
ICML 2023

BLIP-2: Background

Foundation LMs exploded!



[1] Yin, Shukang, et al. "A survey on multimodal large language models." *National Science Review* 11.12 (2024): nwae403.

BLIP-2: Background

Foundation LMs exploded!

- **Great visual encoder** (e.g., CLIP/EVA/ViT variants) already trained on billions of image-text pairs
- **Great language models** with strong reasoning but no direct access to pixels

But...

Scaling multimodal **training** to match these separately-trained giants was **prohibitive**

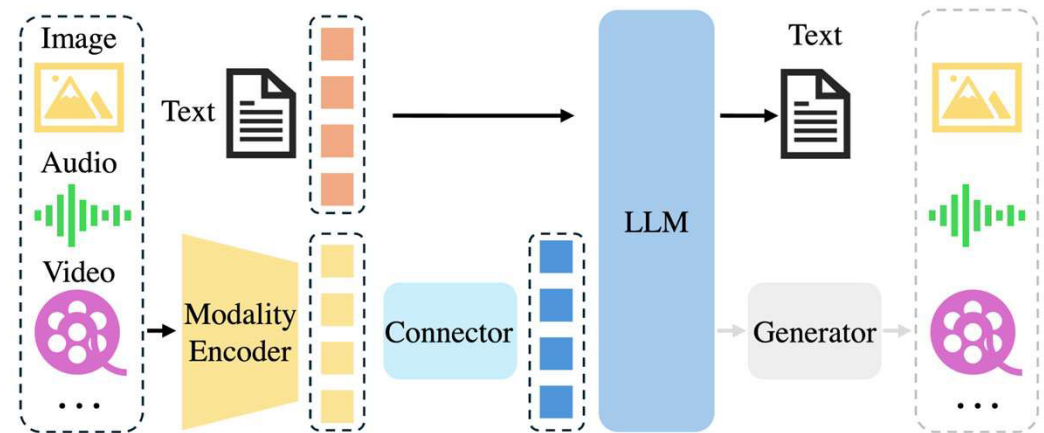


Figure from [1]

BLIP-2: Objective

BLIP2's Question:

Can we “plug” a frozen vision backbone into a frozen LLM and still get state-of-the-art multimodal reasoning, with only a small trainable adapter in the middle?

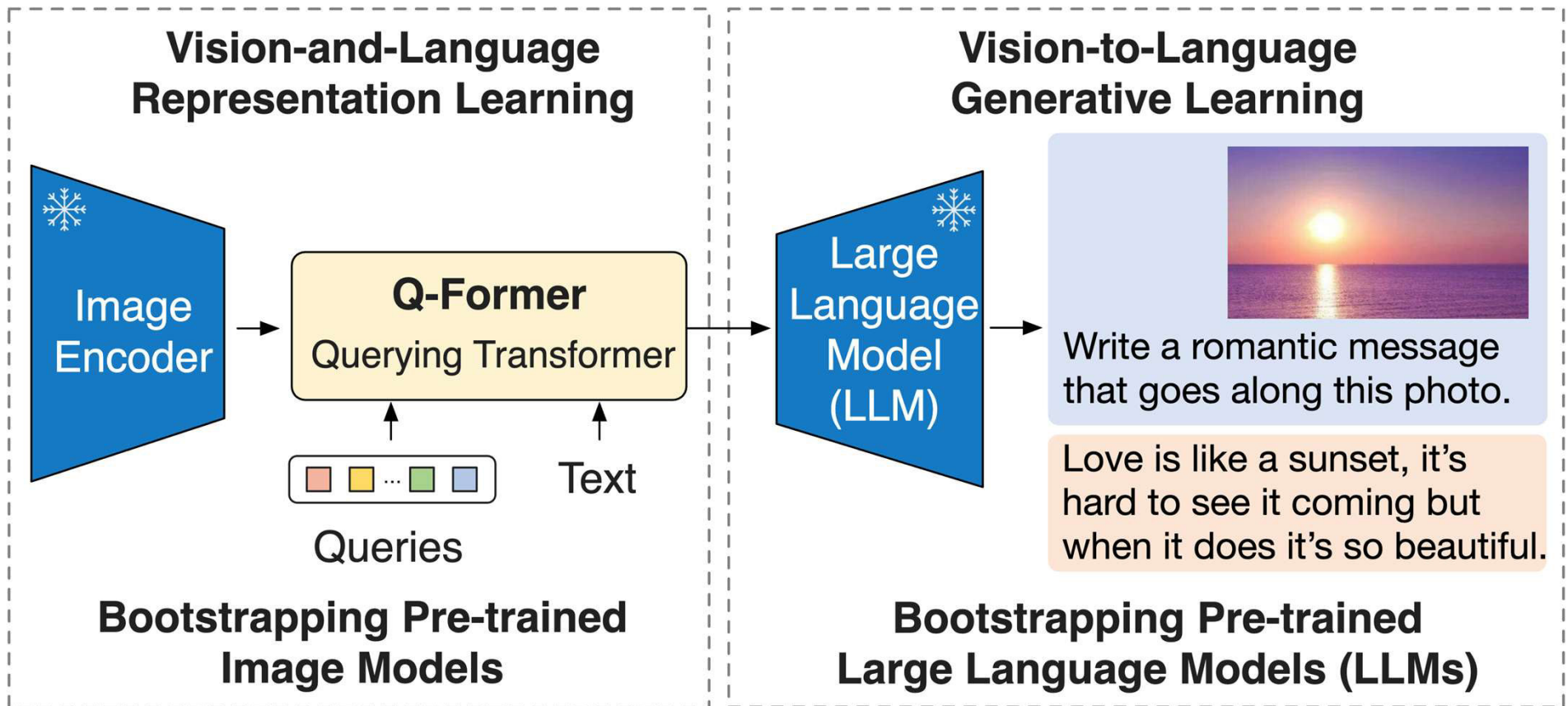
Create an efficient, general bridge between the two

- Reuse the frozen experts without retraining them jointly on massive multimodal corpora

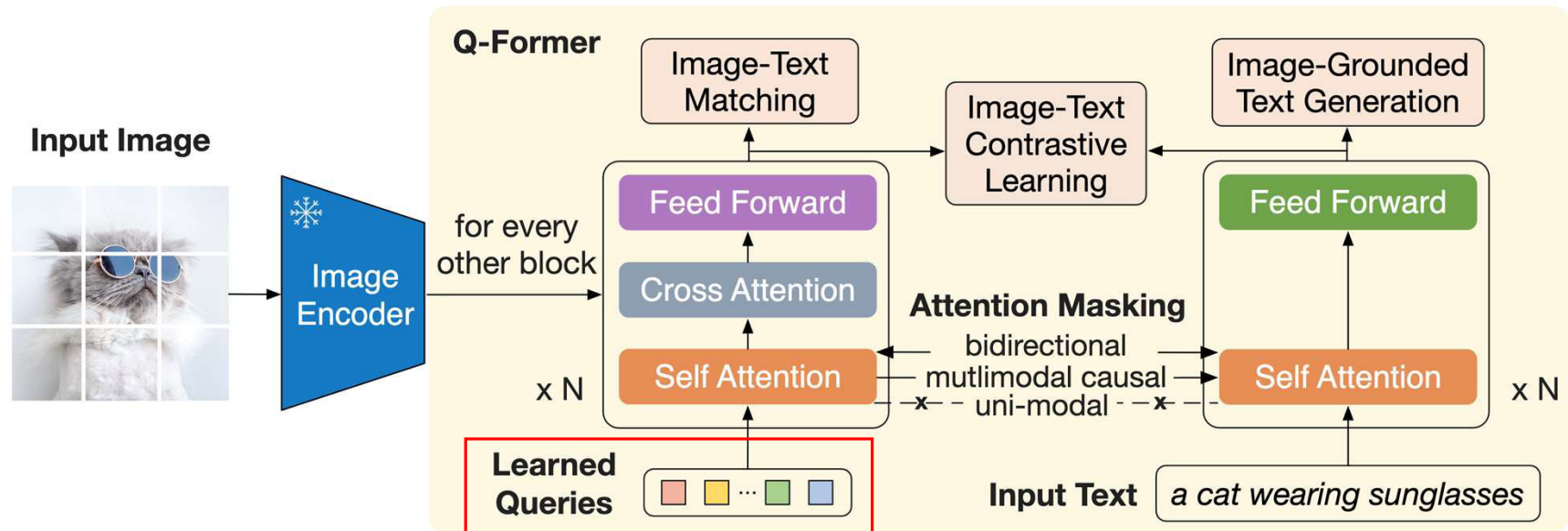
Freeze both giants, train a smart adapter!

BLIP-2: Method

Two-Staged Training and train only the proposed Q-Former

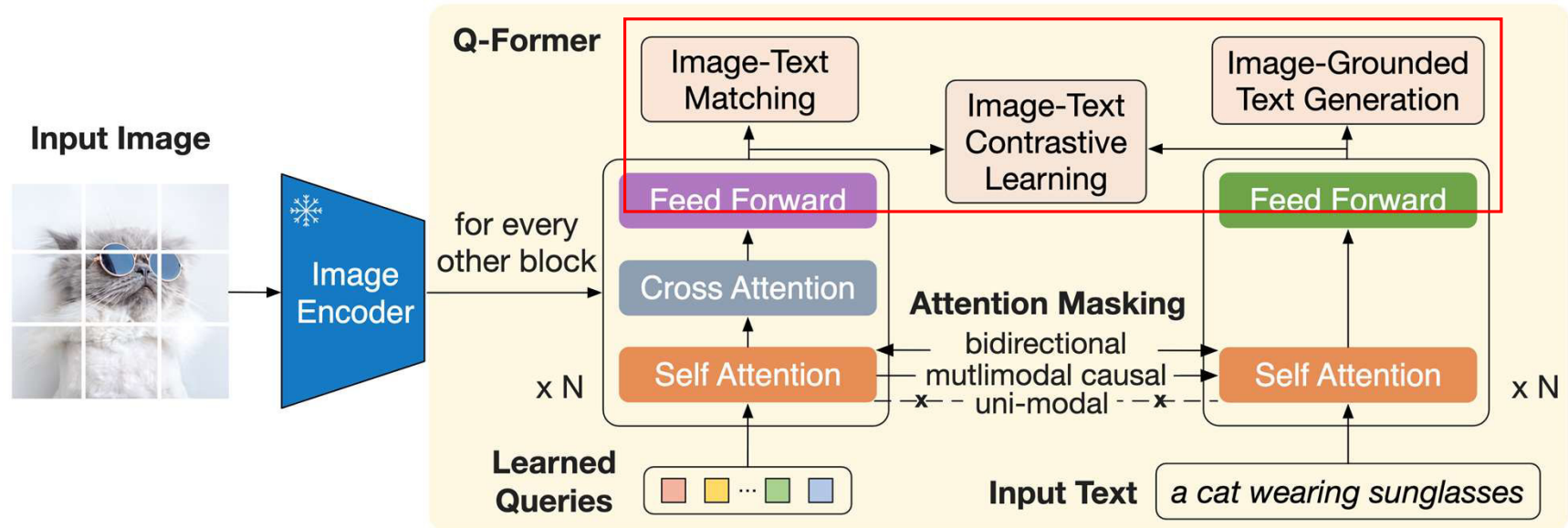


BLIP-2: Method – Stage 1 Training



Learnable Query: small number of learnable query tokens (32-64) that cross-attend to Image/Text tokens

BLIP-2: Method – Stage 1 Training

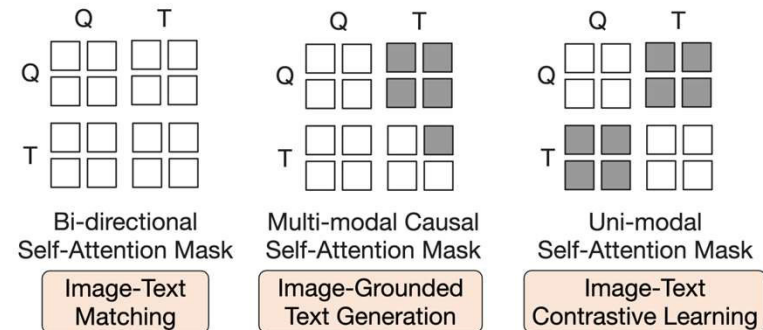


Q: query token positions; **T:** text token positions.

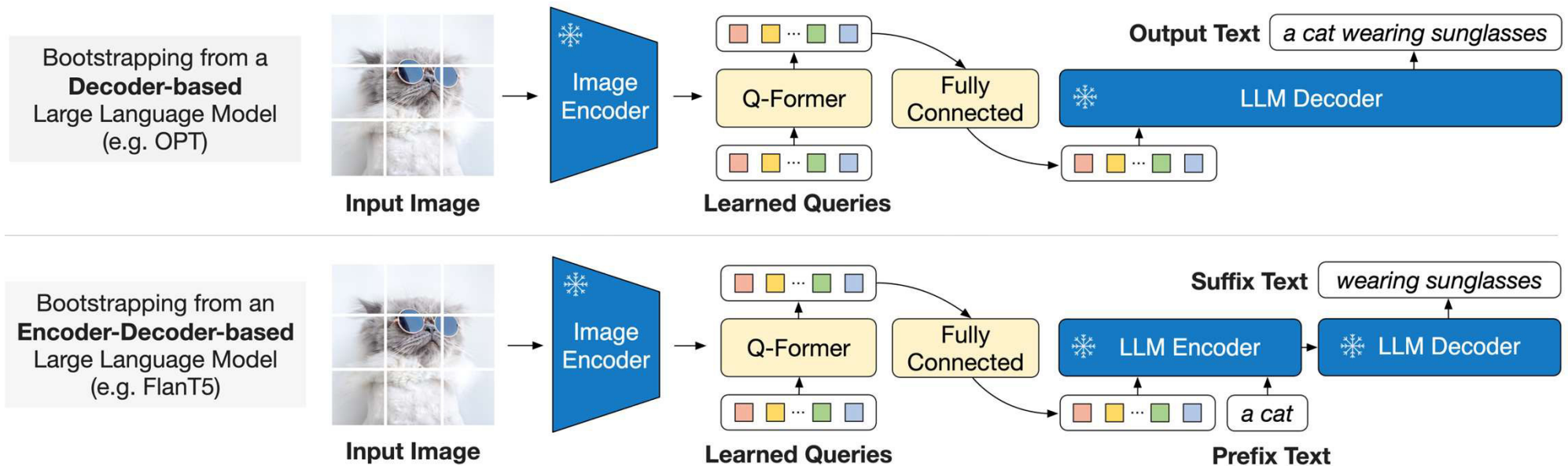
■ masked □ unmasked

Three Pretraining Tasks

Self-attention masking strategy:



BLIP-2: Method – Stage 2 Training



Add trainable fully connected layer to embed the learned queries into LLM space

BLIP-2: Experimental Results

Models	#Trainable Params	Open- sourced?	Visual Question Answering	Image Captioning		Image-Text Retrieval	
			VQAv2 (test-dev) VQA acc.	NoCaps (val) CIDEr	SPICE	Flickr (test) TR@1	IR@1
BLIP (Li et al., 2022)	583M	✓	-	113.2	14.8	96.7	86.7
SimVLM (Wang et al., 2021b)	1.4B	✗	-	112.2	-	-	-
BEIT-3 (Wang et al., 2022b)	1.9B	✗	-	-	-	94.9	81.5
Flamingo (Alayrac et al., 2022)	10.2B	✗	56.3	-	-	-	-
BLIP-2	188M	✓	65.0	121.6	15.8	97.6	89.7

Table 1. Overview of BLIP-2 results on various **zero-shot** vision-language tasks. Compared with previous state-of-the-art models. BLIP-2 achieves the highest zero-shot performance while requiring the least number of trainable parameters during vision-language pre-training.

Improve over state-of-the-art on various benchmarks, while only requiring training a fraction of the parameters that used to require training

BLIP-2: Visualization



Explain the advantages of this product.

The audi e-tron quattro concept is a plug-in hybrid electric sports car that has a range of 310 miles and can accelerate from 0 to 60 mph in just 4 seconds.



Tell me something about the history of this place.

The great wall of china was built in 221 bc by emperor qin shihuang to protect the emperor's capital from invaders from the north.



Write down the facts that you know about this flower.

It is a genus of flowering plants in the orchid family, orchidaceae. It is native to eastern asia and south east asia. It is a perennial flowering plant that grows up to a height of 1 m.



Is this photo unusual?

Yes, it's a house that looks like it's upside down.

How could someone get out of the house?

It has a slide on the side of the house.



What are shown in the photo?

A man and a chicken.

What does the man feel and why?

He is scared of the chicken because it is flying at him.



What are the ingredients I need to make this?

Pepperoni, mozzarella cheese, pizza sauce, olive oil, salt, pepper, basil.

What is the first step?

Place the pizza dough on a baking sheet, brush with olive oil, sprinkle with salt, pepper, and basil.

BLIP-2: Summary

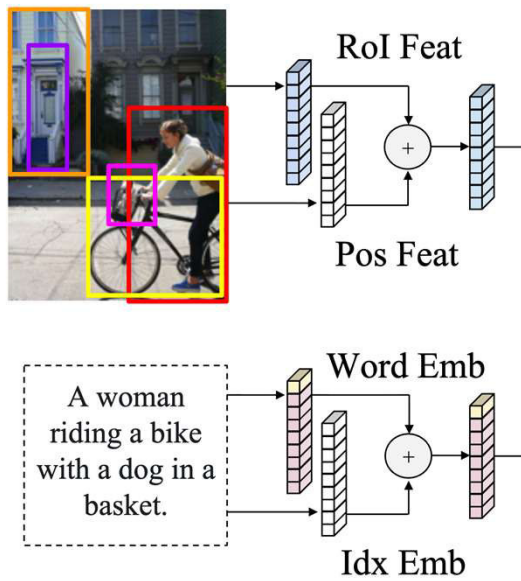
A small, learned query module (Q-Former) can translate pixels into a **compact, language-friendly token set**, letting a **frozen LLM** reason over images without end-to-end retraining.

- Delivers state-of-the-art quality with a fraction of the compute
- Modularizes multimodal systems

Limitations:

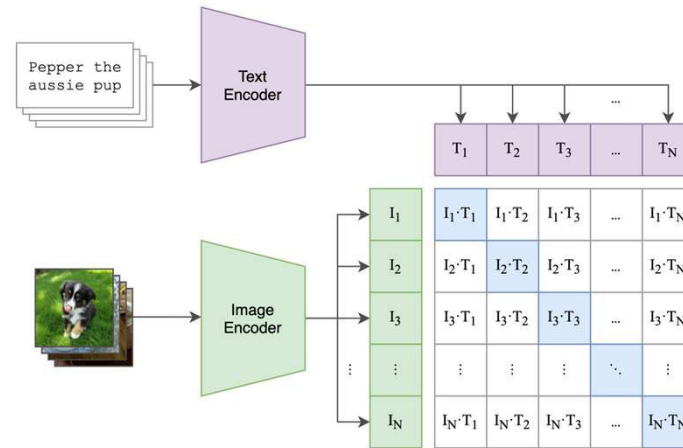
- Fixed bottleneck:
 - **Queries:** great for efficiency but can miss fine spatial details
 - **Frozen LLM:** BLIP2 shines on classic VL benchmarks but ceilings on LLM's frozen knowledge on knowledge-heavy tasks

LXMERT

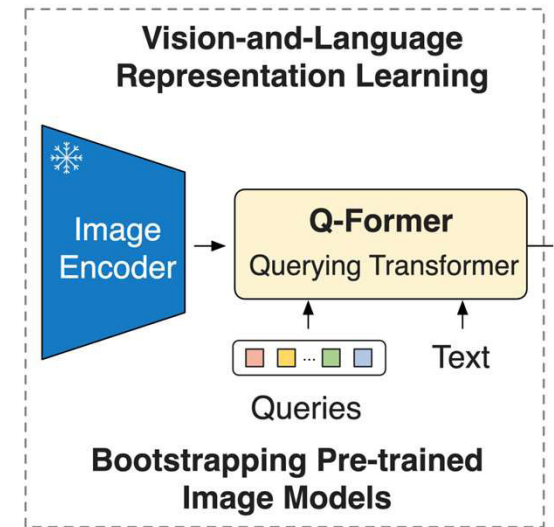


CLIP

(1) Contrastive pre-training



BLIP-2



Thank you!

Vision-Language-Action (VLA) models

Sebastian Joseph

9/25/2025

Vision-Language Models



Explain the advantages of this product.

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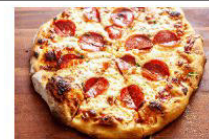


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What is the first step?

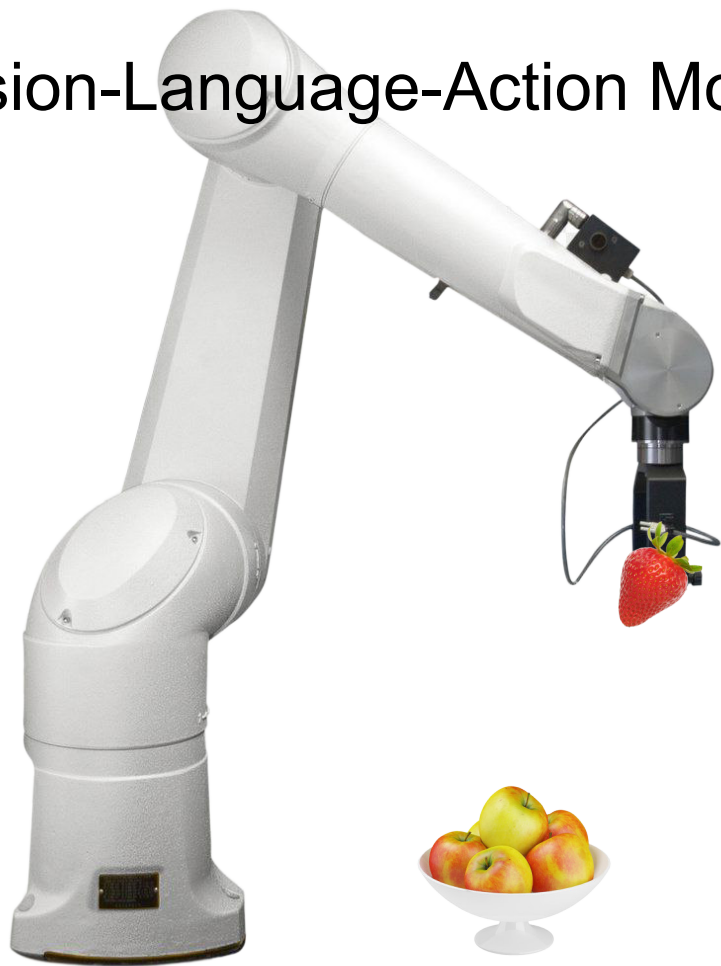
Place the pizza dough on a baking sheet, brush with olive oil, sprinkle with salt, pepper, and basil.

Vision-Language-Action Models

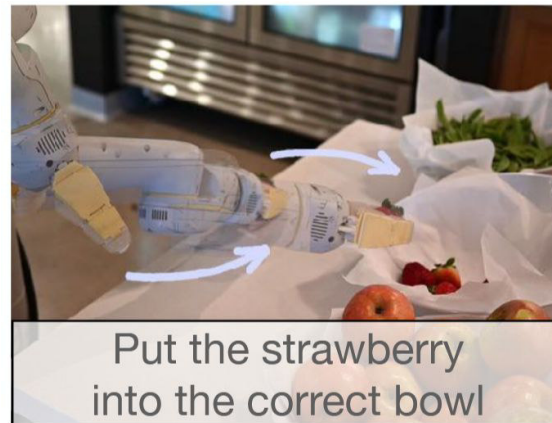


Robots!

Vision-Language-Action Models



Robots!



Vision-Language-Action Models



Not Robots!

Put the strawberry into the bowl

Sure! 🍷 🍓

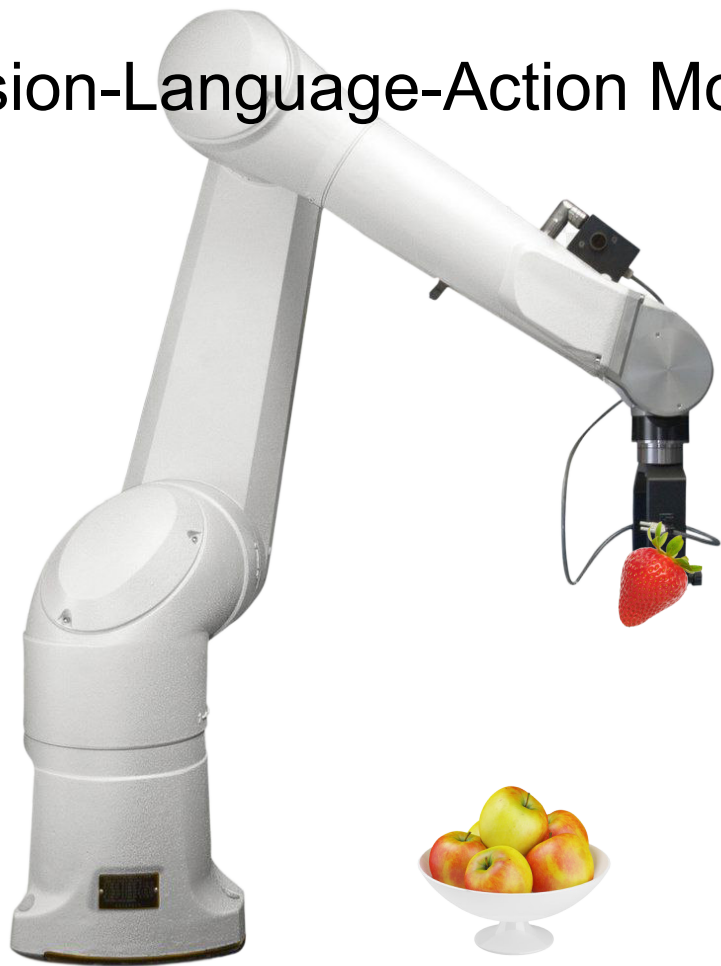
The strawberry is now in the bowl.

Would you like to add anything else?

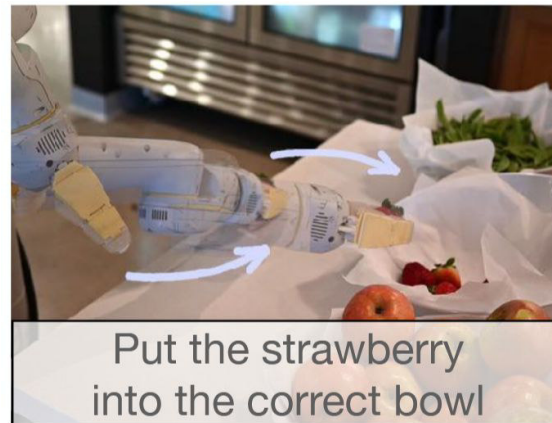


into the correct bowl

Vision-Language-Action Models



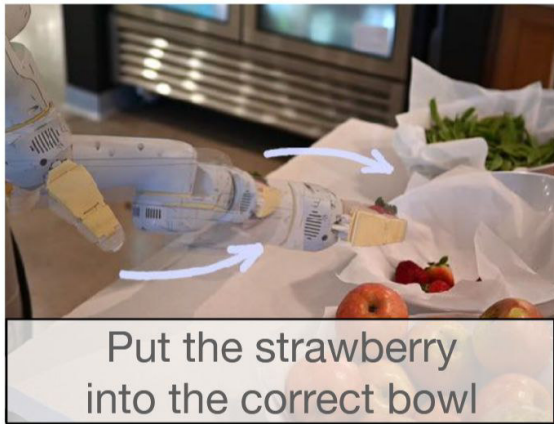
Robots!



Vision-Language-Action Models



Robots!



Decision Transformer: Reinforcement Learning via Sequence Modeling

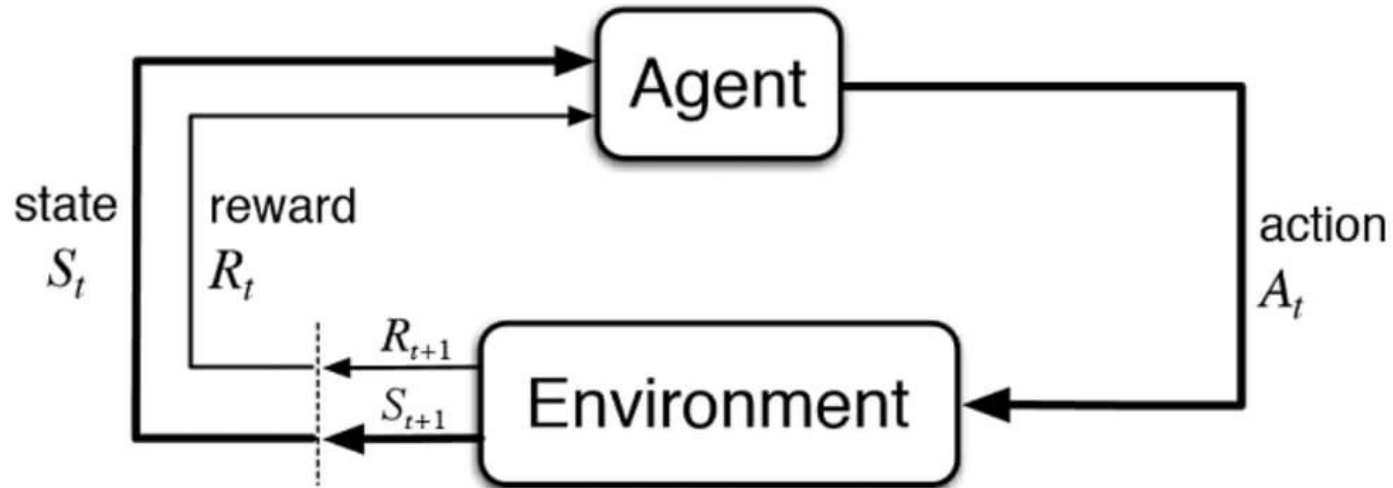
Lili Chen^{*,1}, Kevin Lu^{*,1}, Aravind Rajeswaran², Kimin Lee¹,
Aditya Grover², Michael Laskin¹, Pieter Abbeel¹, Aravind Srinivas^{†,1}, Igor Mordatch^{†,3}

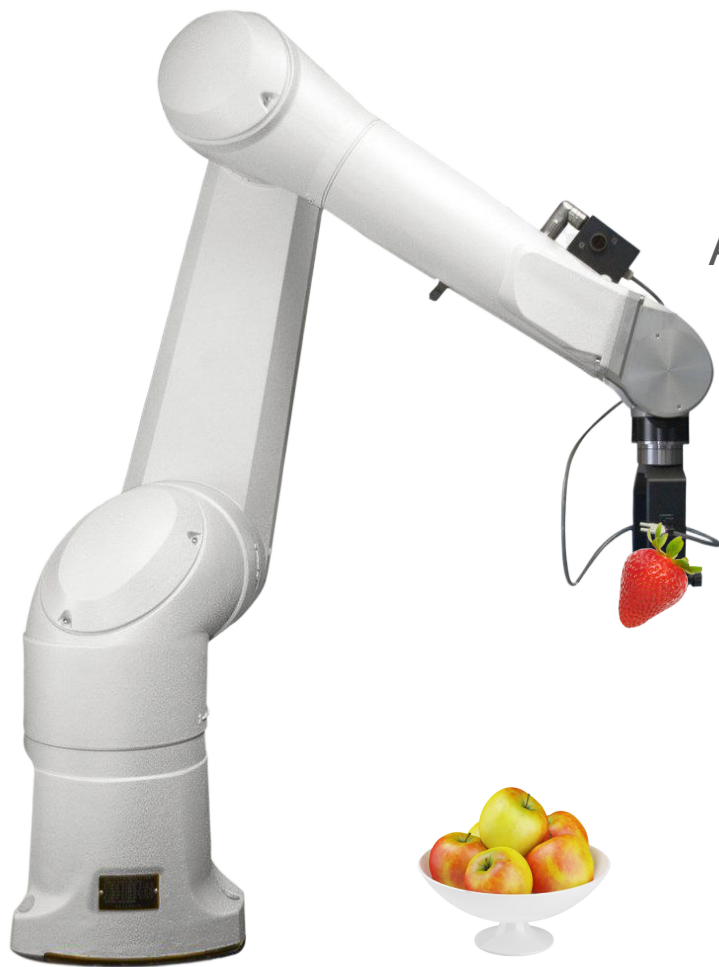
^{*}equal contribution [†]equal advising

¹UC Berkeley ²Facebook AI Research ³Google Brain

{lilichen, kzl}@berkeley.edu

Background: Reinforcement Learning





Agent

*“Make strawberry
apple crumble”*



Environment



Agent

Action



Environment



Agent

Bad Robot!

Reward

Action



Environment



Agent

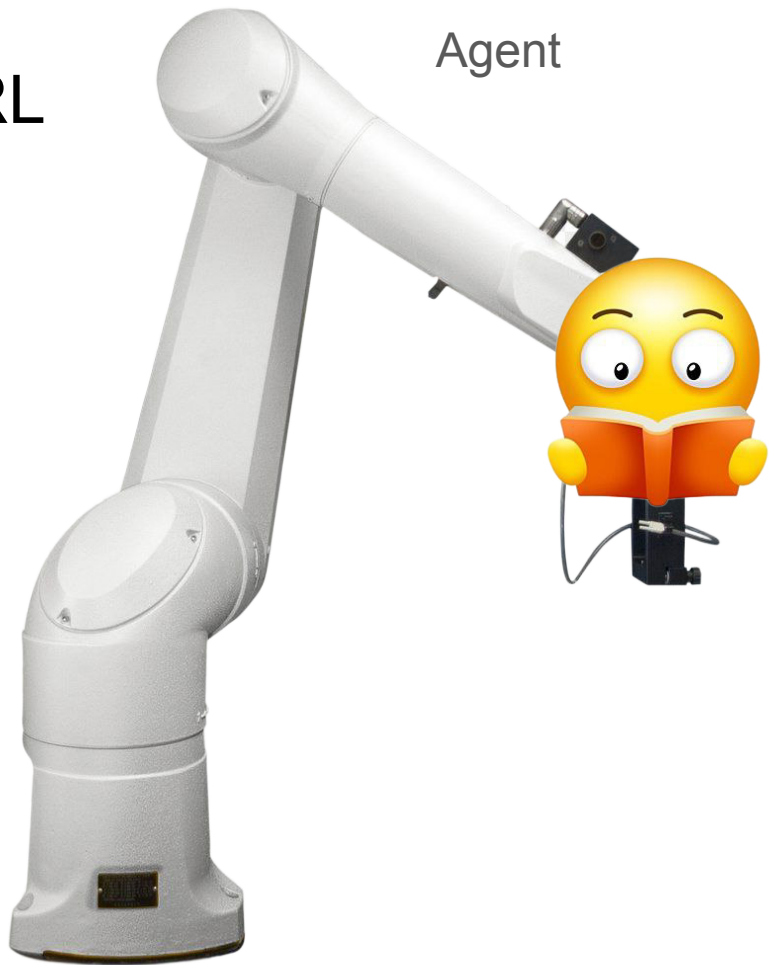
Good Robot! *Reward*

Action

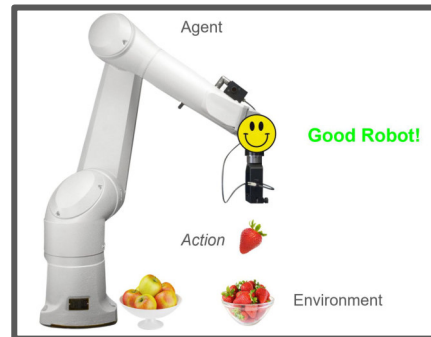


Environment

Offline RL

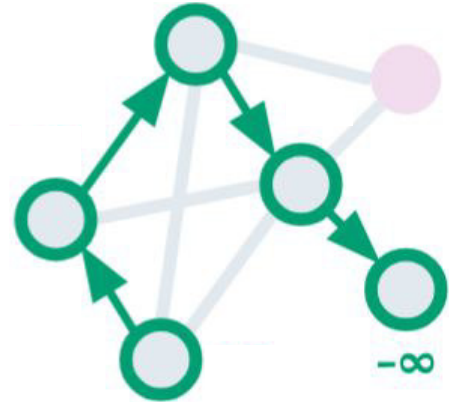
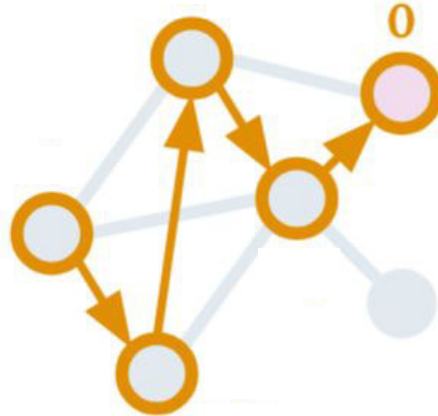
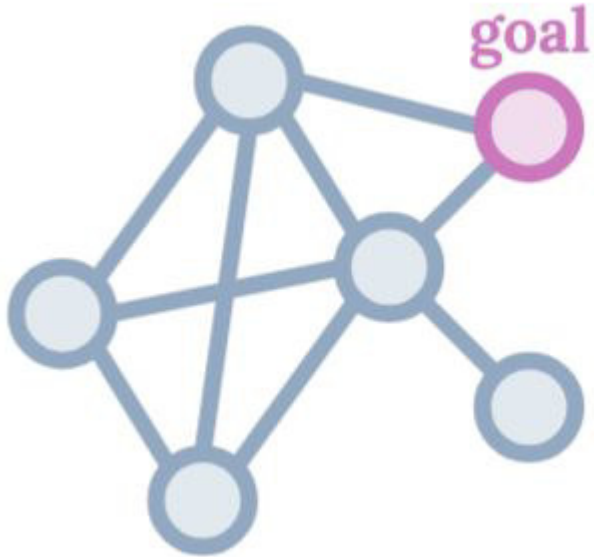


Offline Dataset

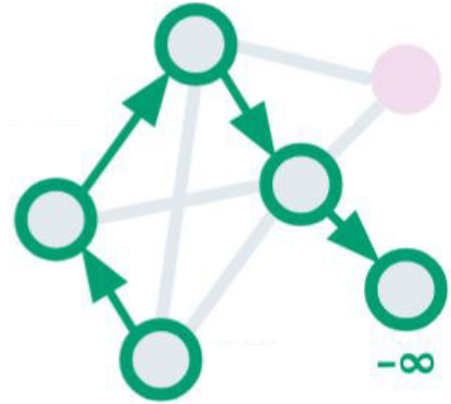
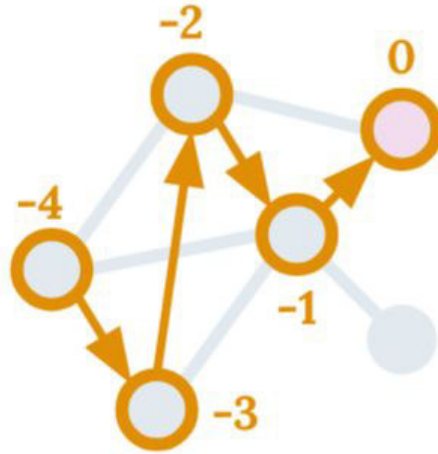
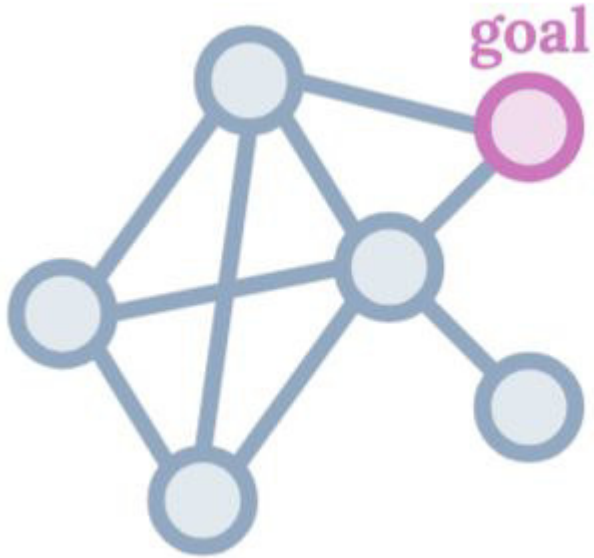


...

Background: Temporal Difference Learning



Background: Temporal Difference Learning

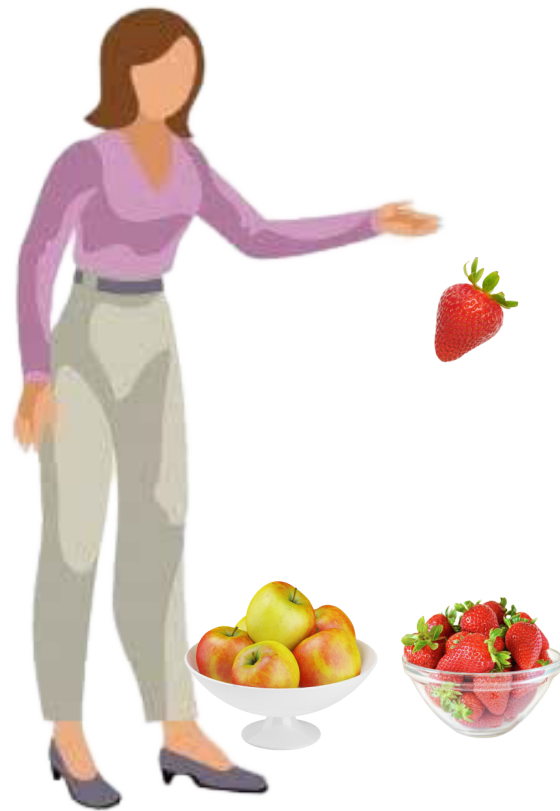
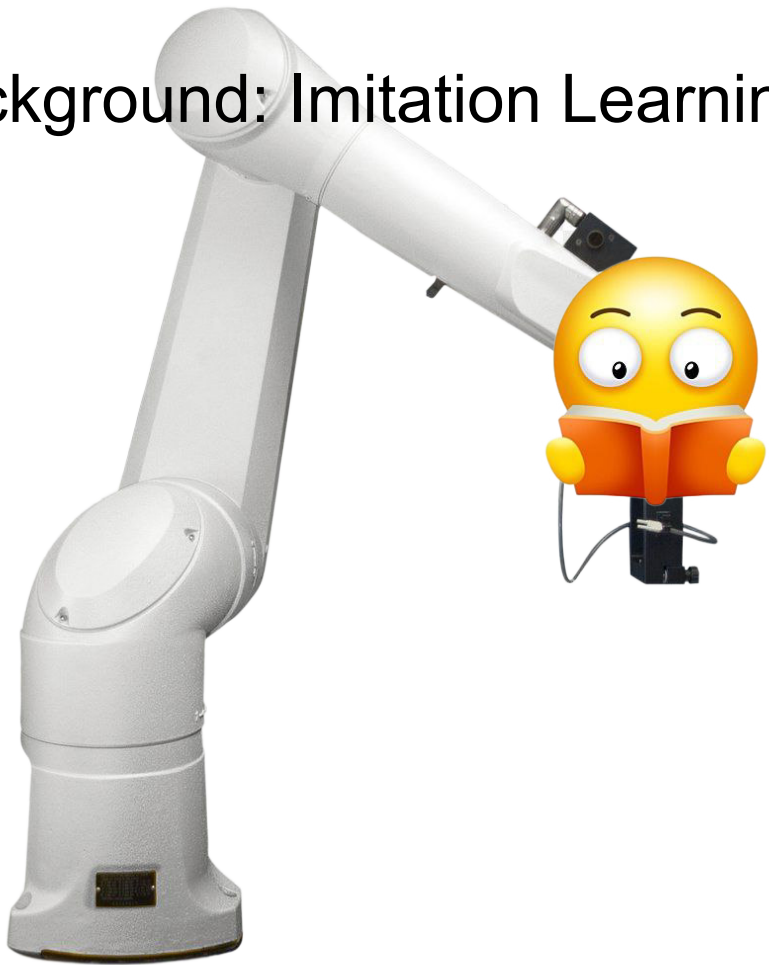


Background: Q-learning & Bellman Equation

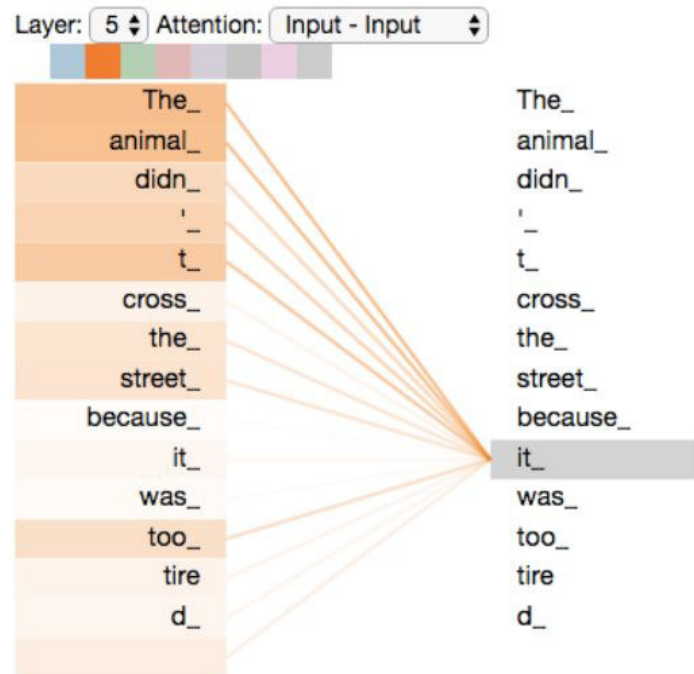
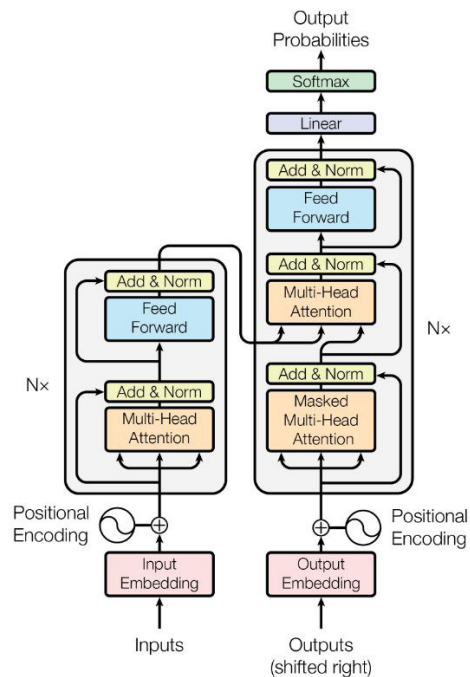
- Q-function -> estimation of function given state and action

$$\underbrace{NewQ(s, a)}_{\text{New Q-Value}} = \underbrace{Q(s, a)}_{\text{Current Q-Value}} + \underbrace{\alpha}_{\text{Learning rate}} [\underbrace{R(s, a)}_{\text{Reward}} + \underbrace{\gamma}_{\text{Discount rate}} \underbrace{\max Q'(s', a')}_{\text{Maximum predicted reward, given new state and all possible actions}} - \underbrace{Q(s, a)}_{\text{Current Q-Value}}]$$

Background: Imitation Learning

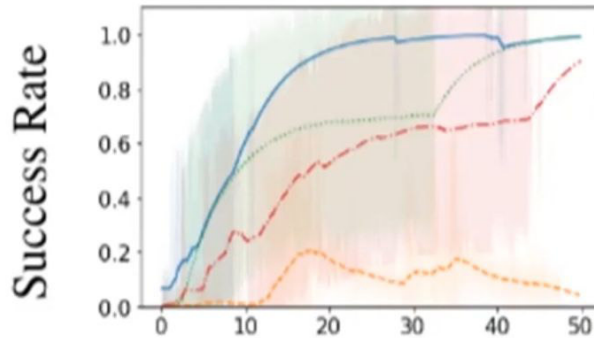


Background: Transformers



Problem

- Unstable training
- “Bootstrapping for long-term credit assignment”
 - Trying to predict long-term rewards into the future
-> error propagation
- “Discounting future rewards”
 - Causes model to think short-term instead of long-term
- LLMs can scale, RL-agents have a harder time



RL -> *generative trajectory modeling*

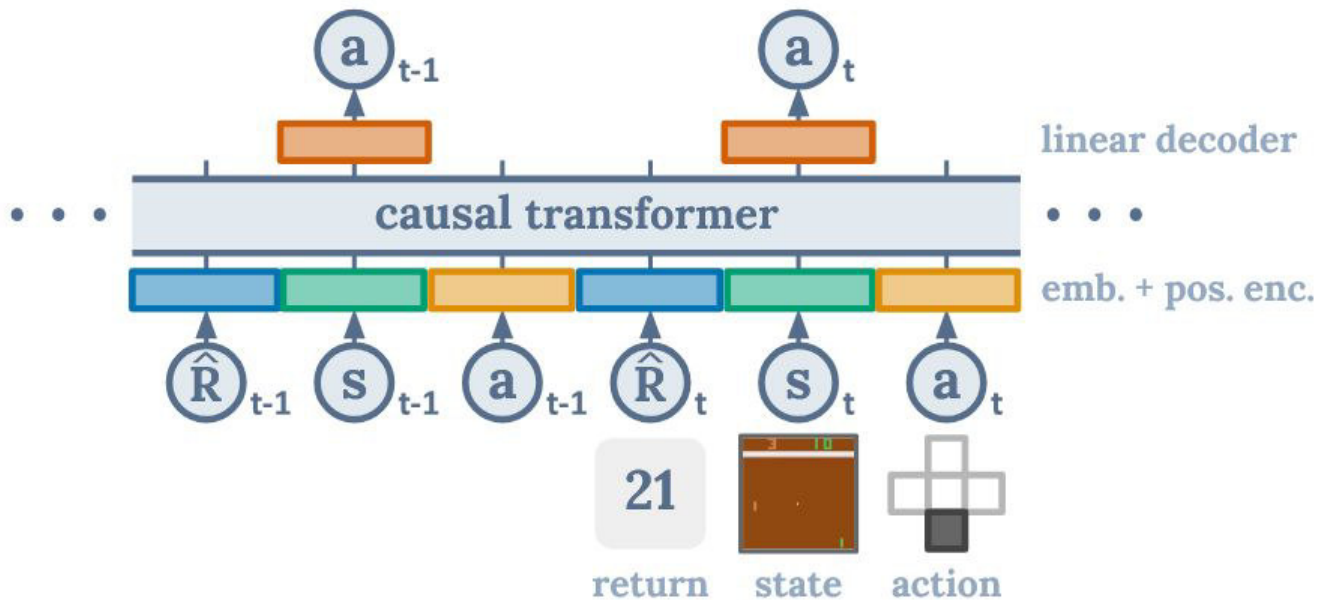
$$\tau = \left(\hat{R}_1, s_1, a_1, \hat{R}_2, s_2, a_2, \dots, \hat{R}_T, s_T, a_T \right)$$

RL -> *generative trajectory modeling*

$$\tau = \left(\hat{R}_1, s_1, a_1, \hat{R}_2, s_2, a_2, \dots, \hat{R}_T, s_T, a_T \right)$$

RL -> ***Sequence Modeling***

Decision Transformer



Decision Transformer

```
# main model
def DecisionTransformer(R, s, a, t):
    # compute embeddings for tokens
    pos_embedding = embed_t(t) # per-timestep (note: not per-token)
    s_embedding = embed_s(s) + pos_embedding
    a_embedding = embed_a(a) + pos_embedding
    R_embedding = embed_R(R) + pos_embedding

    # interleave tokens as (R_1, s_1, a_1, ..., R_K, s_K)
    input_embeds = stack(R_embedding, s_embedding, a_embedding)

    # use transformer to get hidden states
    hidden_states = transformer(input_embeds=input_embeds)

    # select hidden states for action prediction tokens
    a_hidden = unstack(hidden_states).actions

    # predict action
    return pred_a(a_hidden)
```

3 tokens - same time step

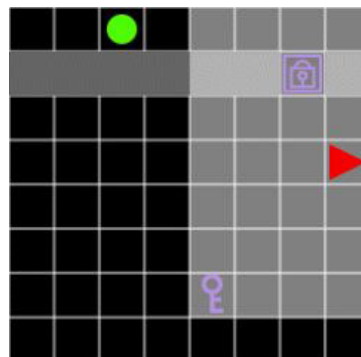
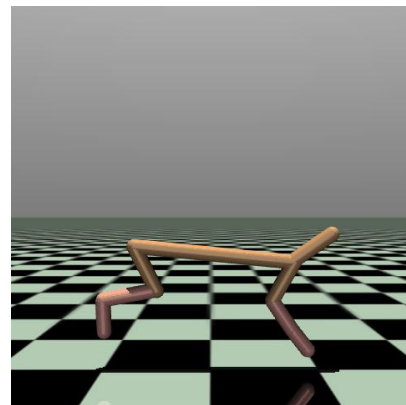
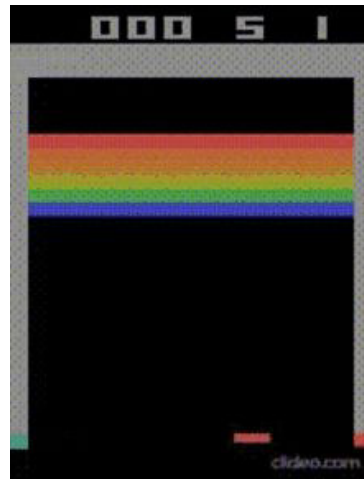
Training DT

```
# training loop
for (R, s, a, t) in dataloader: # dims: (batch_size, K, dim)
    a_preds = DecisionTransformer(R, s, a, t)
    loss = mean((a_preds - a)**2) # L2 loss for continuous actions
    optimizer.zero_grad(); loss.backward(); optimizer.step()
```

Cross-entropy loss for discrete
actions

Evaluation

- Constant Baseline 1: **Temporal Difference Learning**
 - **Conservative Q-Learning**
- Constant Baseline 2: **Imitation Learning**
 - Behavior cloning - essentially supervised learning
- Benchmark 1: **Atari**
 - Normalized-gamer scores
 - 1% of DQN-replay dataset -> 500k examples
- Benchmark 2: **OpenAI Gym**
 - Scores are normalized so 100 represents equivalence to expert
 - 3 Datasets: Medium, Medium-Replay, Medium-Expert
- Task 3: **Key-to-Door**
 - Success Rate



Overall Results

- DT performs comparably or better than baselines
- Especially long-term tasks

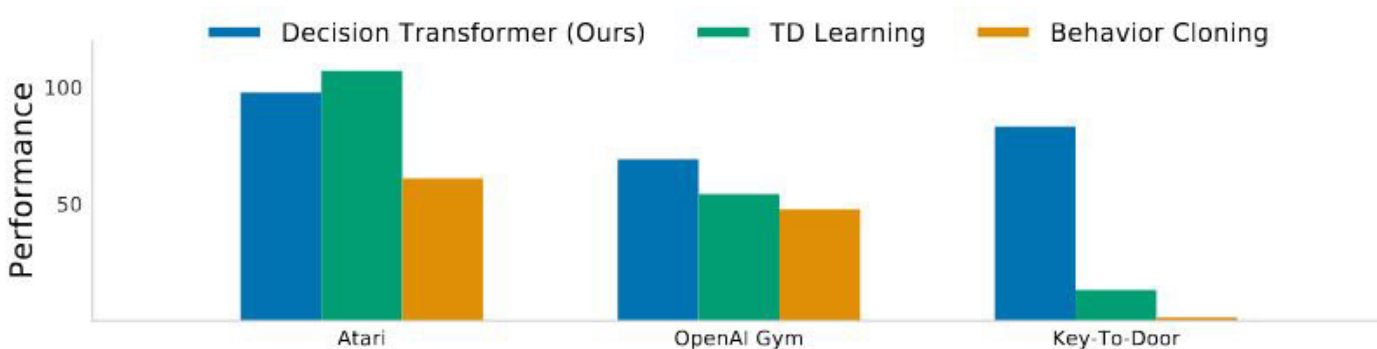


Figure 3: Results comparing Decision Transformer (ours) to TD learning (CQL) and behavior cloning across Atari, OpenAI Gym, and Minigrid. On a diverse set of tasks, Decision Transformer performs comparably or better than traditional approaches. Performance is measured by normalized episode return (see text for details).

Decision Transformer = Behavior Cloning?

When data is plentiful -> BC beats DT

Dataset	Environment	DT (Ours)	10%BC	25%BC	40%BC	100%BC	CQL
Medium	HalfCheetah	42.6 $\pm$ 0.1	42.9	43.0	43.1	43.1	44.4
Medium	Hopper	67.6 $\pm$ 1.0	65.9	65.2	65.3	63.9	58.0
Medium	Walker	74.0 $\pm$ 1.4	78.8	80.9	78.8	77.3	79.2
Medium	Reacher	51.2 $\pm$ 3.4	51.0	48.9	58.2	58.4	26.0
Medium-Replay	HalfCheetah	36.6 $\pm$ 0.8	40.8	40.9	41.1	4.3	46.2
Medium-Replay	Hopper	82.7 $\pm$ 7.0	70.6	58.6	31.0	27.6	48.6
Medium-Replay	Walker	66.6 $\pm$ 3.0	70.4	67.8	67.2	36.9	26.7
Medium-Replay	Reacher	18.0 $\pm$ 2.4	33.1	16.2	10.7	5.4	19.0
Average		56.1	56.7	52.7	49.4	39.5	43.5

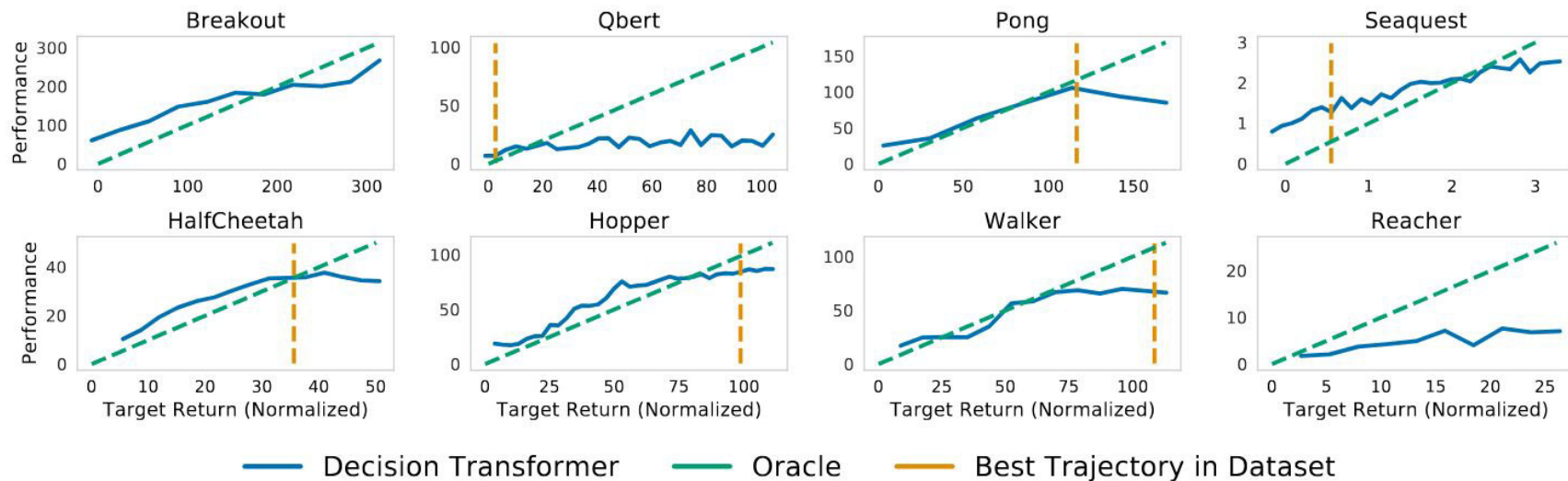
Table 3: Comparison between Decision Transformer (DT) and Percentile Behavior Cloning (%BC).

Decision Transformer = Behavior Cloning?

When its not -> DT wins

Game	DT (Ours)	10%BC	25%BC	40%BC	100%BC
Breakout	267.5 $\pm$ 97.5	28.5 $\pm$ 8.2	73.5 $\pm$ 6.4	108.2 $\pm$ 67.5	138.9 $\pm$ 61.7
Qbert	15.4 $\pm$ 11.4	6.6 $\pm$ 1.7	16.0 $\pm$ 13.8	11.8 $\pm$ 5.8	17.3 $\pm$ 14.7
Pong	106.1 $\pm$ 8.1	2.5 $\pm$ 0.2	13.3 $\pm$ 2.7	72.7 $\pm$ 13.3	85.2 $\pm$ 20.0
Seaquest	2.5 $\pm$ 0.4	1.1 $\pm$ 0.2	1.1 $\pm$ 0.2	1.6 $\pm$ 0.4	2.1 $\pm$ 0.3

Can DT model returns?



Context Length is Needed

Game	DT (Ours)	DT with no context ($K = 1$)
Breakout	267.5 ± 97.5	73.9 ± 10
Qbert	15.1 ± 11.4	13.6 ± 11.3
Pong	106.1 ± 8.1	2.5 ± 0.2
Seaquest	2.5 ± 0.4	0.6 ± 0.1

Long-term Credit Assignment

Dataset	DT (Ours)	CQL	BC	%BC	Random
1K Random Trajectories	71.8%	13.1%	1.4%	69.9%	3.1%
10K Random Trajectories	94.6%	13.3%	1.6%	95.1%	3.1%

Table 6: Success rate for Key-to-Door environment. Methods using hindsight (Decision Transformer, %BC) can learn successful policies, while TD learning struggles to perform credit assignment.

Sparse Reward Setting

Dataset	Environment	Delayed (Sparse)		Agnostic		Original (Dense)	
		DT (Ours)	CQL	BC	%BC	DT (Ours)	CQL
Medium-Expert	Hopper	107.3 ± 3.5	9.0	59.9	102.6	107.6	111.0
Medium	Hopper	60.7 ± 4.5	5.2	63.9	65.9	67.6	58.0
Medium-Replay	Hopper	78.5 ± 3.7	2.0	27.6	70.6	82.7	48.6

Table 7: Results for D4RL datasets with delayed (sparse) reward. Decision Transformer (DT) and imitation learning are minimally affected by the removal of dense rewards, while CQL fails.

To Summarize

- RL can be modeled as a sequence modeling objective
 - This is not the same as behavior cloning; they are learning from intermediate rewards
 - Self-attention enables long-term credit assignment
- Self-attention also enables a sparse reward setting
- Now we can scale

RT-1: ROBOTICS TRANSFORMER FOR REAL-WORLD CONTROL AT SCALE

¹ Anthony Brohan*, Noah Brown*, Justice Carbajal*, Yevgen Chebotar*, Joseph Dabis*, Chelsea Finn*, Keerthana Gopalakrishnan*, Karol Hausman*, Alex Herzog[†], Jasmine Hsu*, Julian Ibarz*, Brian Ichter*, Alex Irpan*, Tomas Jackson*, Sally Jesmonth*, Nikhil J Joshi*, Ryan Julian*, Dmitry Kalashnikov*, Yuheng Kuang*, Isabel Leal*, Kuang-Huei Lee[‡], Sergey Levine*, Yao Lu*, Utsav Malla*, Deeksha Manjunath*, Igor Mordatch[†], Ofir Nachum[‡], Carolina Parada*, Jodilyn Peralta*, Emily Perez*, Karl Pertsch*, Jornell Quiambao*, Kanishka Rao*, Michael Ryoo*, Grecia Salazar*, Pannag Sanketi*, Kevin Sayed*, Jaspiar Singh*, Sumedh Sontakke[‡], Austin Stone*, Clayton Tan*, Huong Tran*, Vincent Vanhoucke*, Steve Vega*, Quan Vuong*, Fei Xia*, Ted Xiao*, Peng Xu*, Sichun Xu*, Tianhe Yu*, Brianna Zitkovich*

*Robotics at Google, [†]Everyday Robots, [‡]Google Research, Brain Team

Multi-Task Backbone Model for Robotic Tasks

- Getting robotics data is hard
- Trend was on training models on large highly specialized datasets
- What if we had a general model that is more capable of doing everything?
- **Can we make an LLM for robots?**

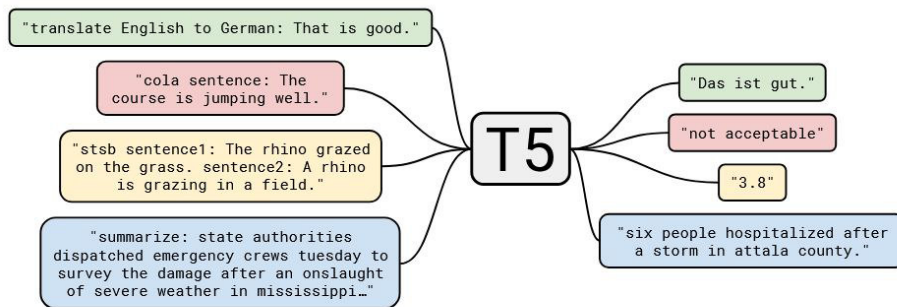
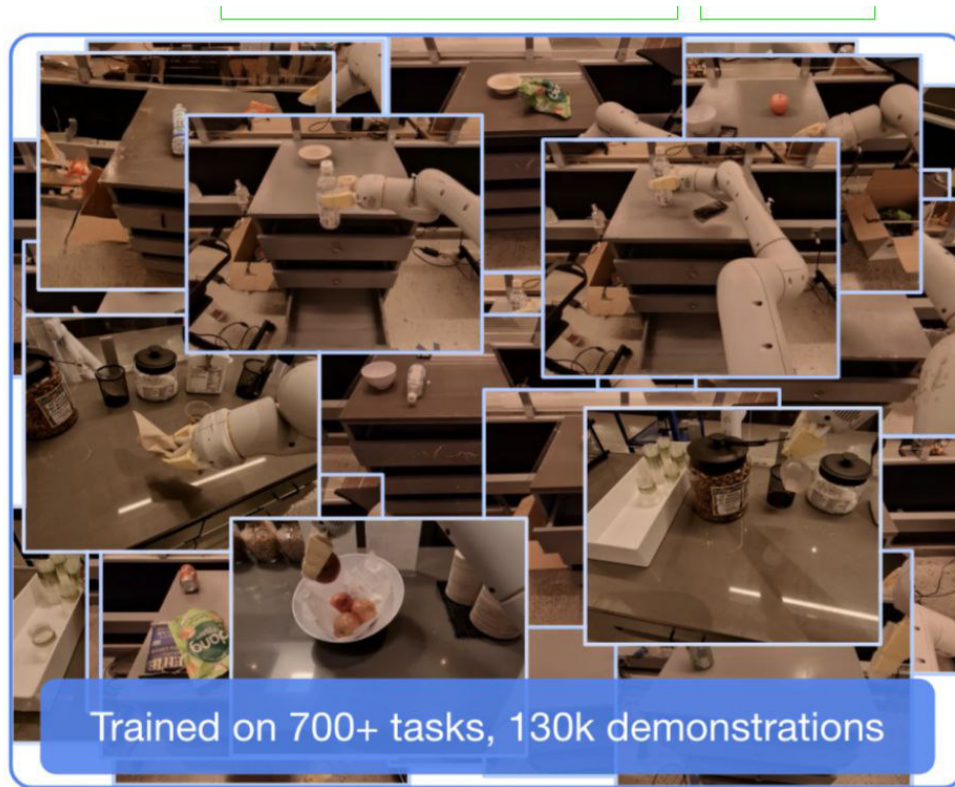


Figure 1: A diagram of our text-to-text framework. Every task we consider—including translation, question answering, and classification—is cast as feeding our model text as input and training it to generate some target text. This allows us to use the same model, loss function, hyperparameters, etc. across our diverse set of tasks. It also provides a standard testbed for the methods included in our empirical survey. “T5” refers to our model, which we dub the “Text-to-Text Transfer Transformer”.

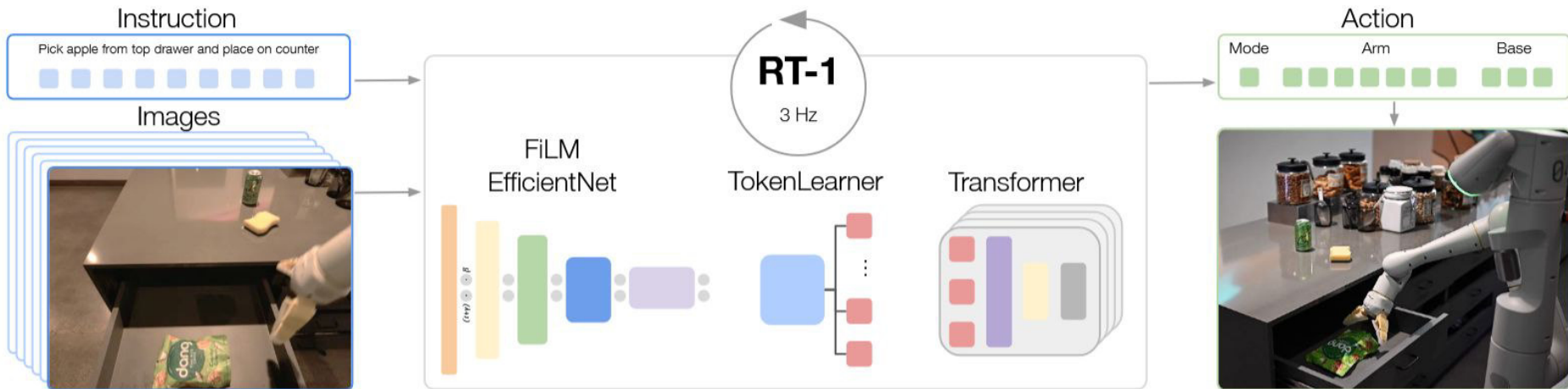
Data Collection

- We'll do the hard work for you
- 130k episodes over more than 700 tasks
- Took 17 months with 13 robots



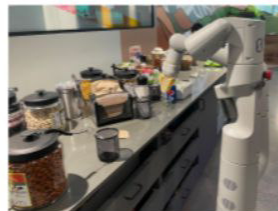
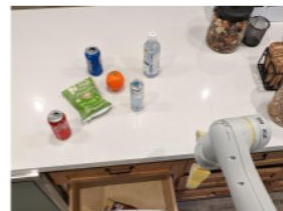
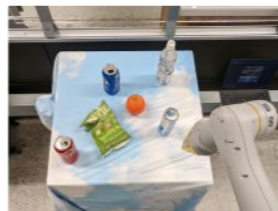
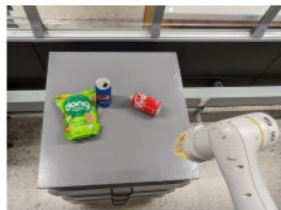
RT-1

- Focus on efficiency
- Trained from scratch
- Imitation Learning



Evaluation

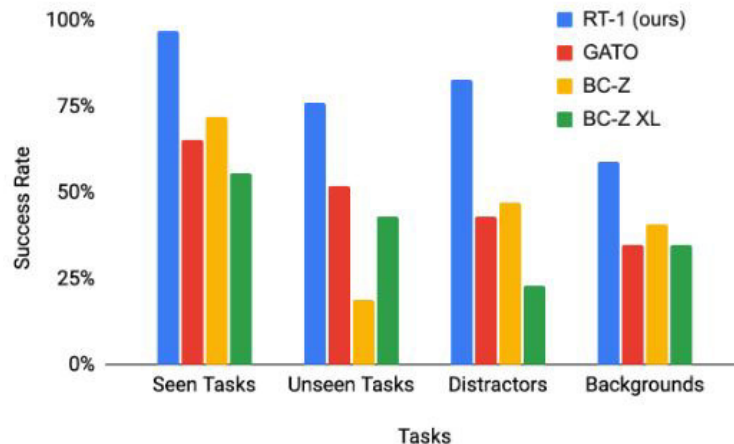
- Evaluate Robots using RT-1 on tasks
- Baselines
 - Gato, BC-Z, BC-Z XL
- Seen tasks
- Unseen tasks
- Robustness tests
 - Robust to Distractors
 - Robust to Background Changes



Can RT-1 perform well and generalize?

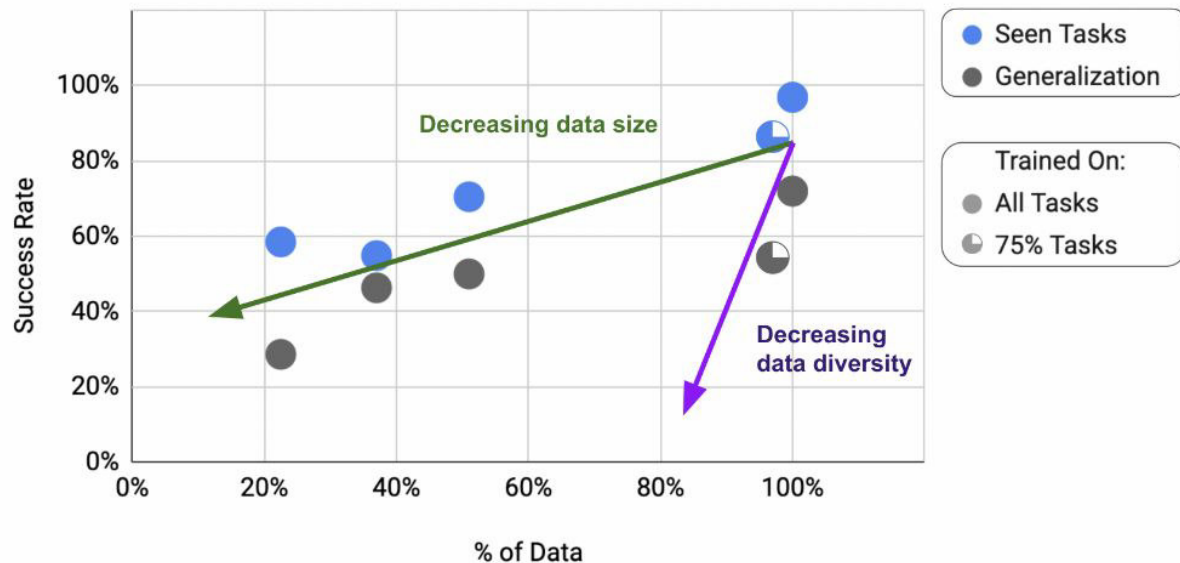
RT-1 outperforms other generalist models; power of data

Model	Seen Tasks	Unseen Tasks	Distractors	Backgrounds
Gato (Reed et al., 2022)	65	52	43	35
BC-Z (Jang et al., 2021)	72	19	47	41
BC-Z XL	56	43	23	35
RT-1 (ours)	97	76	83	59



What is more important: Data Diversity or Data Size?

Data diversity affects performance a lot more than size



Limitations & Analysis

- Imitation learning -> cannot surpass performance of performers
- Cannot generalize to entirely new motion
- Dataset consists of limited set of tasks
- Not really that robust to backgrounds
- **Data diversity matters**

RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, Pete Florence, Chuyuan Fu, Montse Gonzalez Arenas, Keerthana Gopalakrishnan, Kehang Han, Karol Hausman, Alexander Herzog, Jasmine Hsu, Brian Ichter, Alex Irpan, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Lisa Lee, Tsang-Wei Edward Lee, Sergey Levine, Yao Lu, Henryk Michalewski, Igor Mordatch, Karl Pertsch, Kanishka Rao, Krista Reymann, Michael Ryoo, Grecia Salazar, Pannag Sanketi, Pierre Sermanet, Jaspiar Singh, Anikait Singh, Radu Soricut, Huong Tran, Vincent Vanhoucke, Quan Vuong, Ayzaan Wahid, Stefan Welker, Paul Wohlhart, Jialin Wu, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich

Google DeepMind. Authors listed in alphabetical order, with contributions listed in Appendix A.

Data Bottleneck

- RT-1 had several limitations despite such a large, expensive dataset
- LLMs leveraged web knowledge to improve capabilities
- Can we do the same with end-to-end robotic learning?



Trained on 700+ tasks, 130k demonstrations

RT-2

Internet-Scale VQA + Robot Action Data



Q: What is happening in the image?

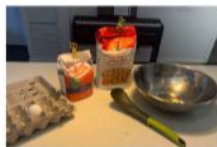
A: 311 423 170 55 244

A grey donkey walks down the street.

Q: Que puis-je faire avec ces objets?

A: 3455 1144 189 25673

Faire cuire un gâteau.



Q: What should the robot do to **<task>**?

A: 132 114 128 5 25 156

$\Delta \text{Translation} = [0.1, -0.2, 0]$
 $\Delta \text{Rotation} = [10^\circ, 25^\circ, -7^\circ]$

Vision-Language-Action Models for Robot Control

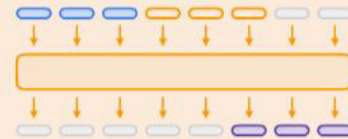
Q: What should the robot do to **<task>**? A: ...



RT-2

ViT

Large Language Model



A: 132 114 128 5 25 156

De-Tokenize

$\Delta T = [0.1, -0.2, 0]$
 $\Delta R = [10^\circ, 25^\circ, -7^\circ]$

Robot Action

Co-Fine-Tune

Deploy

Closed-Loop Robot Control



Put the strawberry into the correct bowl



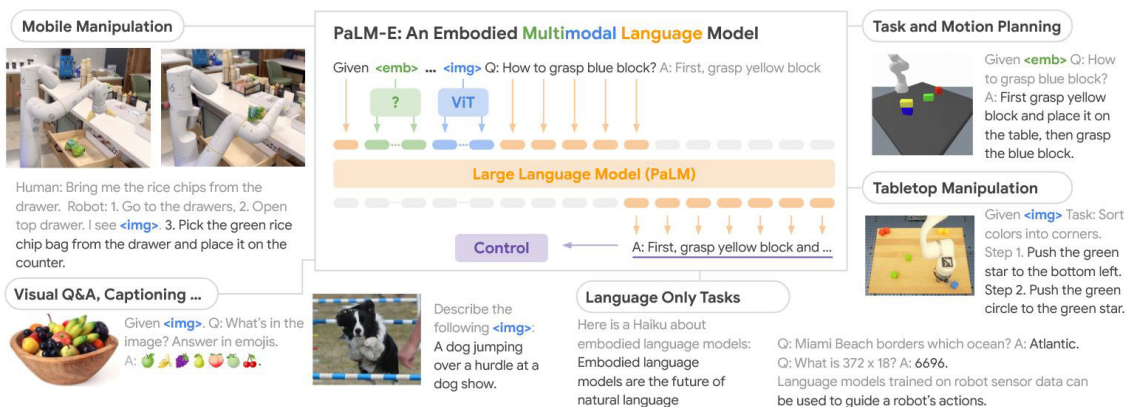
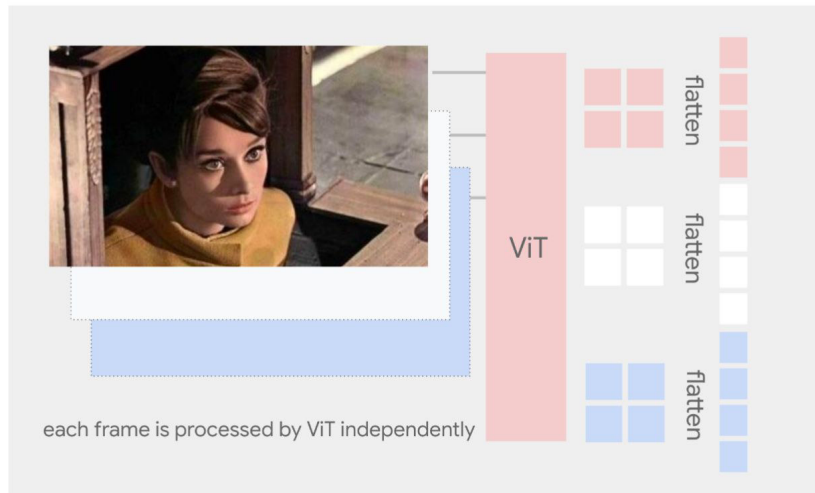
Pick the nearly falling bag



Pick object that is different

Pre-trained VLMs

- PaLI-X
 - Takes ViT-22B and UL2 20B and further train on vision-language mixture data
- PaLM-E
 - 540B PaLM + 22B ViT
 - Trained on embodied data
 - Planning
 - Manipulation
- RT-2
 - Largest models for robotic control



Robot-Action Fine-Tuning

- Action Space

“terminate Δpos_x Δpos_y Δpos_z
 Δrot_x Δrot_y Δrot_z gripper_extension”.

- For PaLI-X use integers, for PaLM use 256 least used tokens.
- Co-fine-tuning with Web data helps model learn Robot Actions
- For output, can only sample action tokens
- Data:
 - Web: PaLI-X, PaLM-E
 - Robot: RT-1



Q: What is happening in the image?

A: 311 423 170 55 244

A grey donkey walks down the street.

Q: Que puis-je faire avec ces objets?

A: 3455 1144 189 25673

Faire cuire un gâteau.



Q: What should the robot do to **<task>**?

A: 132 114 128 5 25 156

$\Delta\text{Translation} = [0.1, -0.2, 0]$

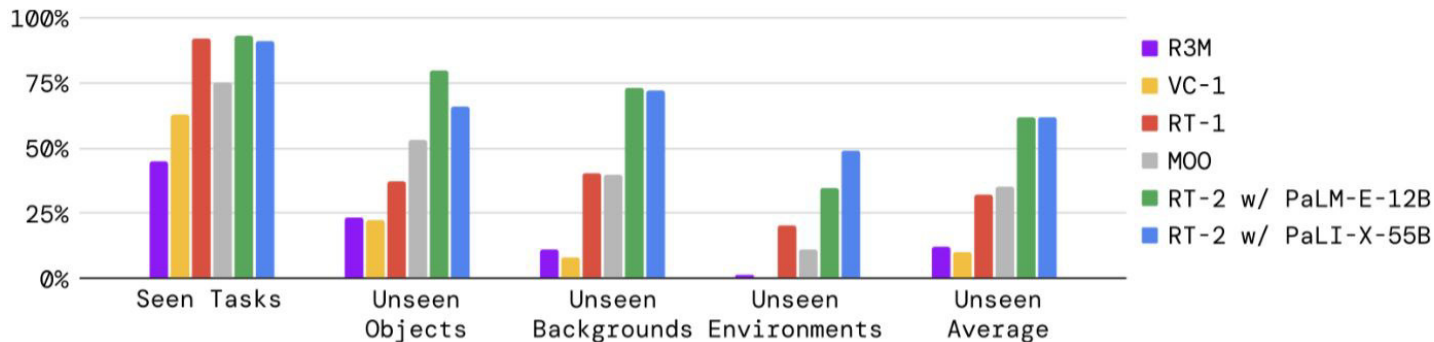
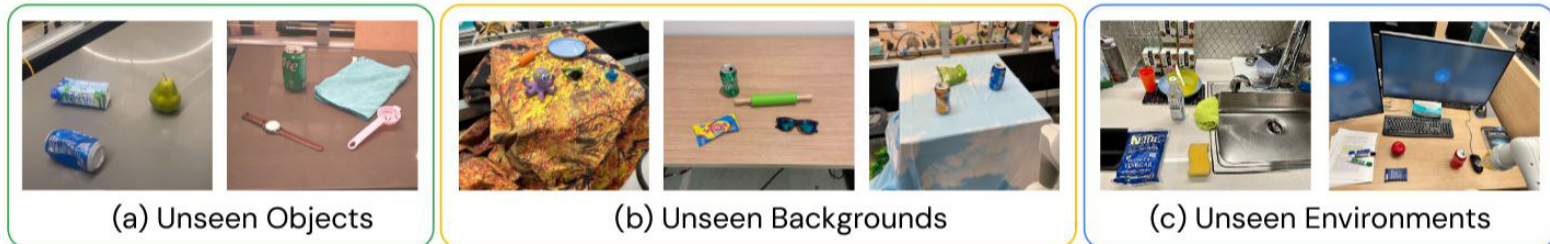
$\Delta\text{Rotation} = [10^\circ, 25^\circ, -7^\circ]$

Baselines

- RM-3
 - Visual representation model
 - Image embeddings pass with language embeddings through token learner
 - Unfreeze weights during training
- VC-1
 - Another visual embedding model
 - Same methodology as above
- MOO
 - Image is marked with object of interest
 - Model is trained with end-to-end policy afterwards
- RT1

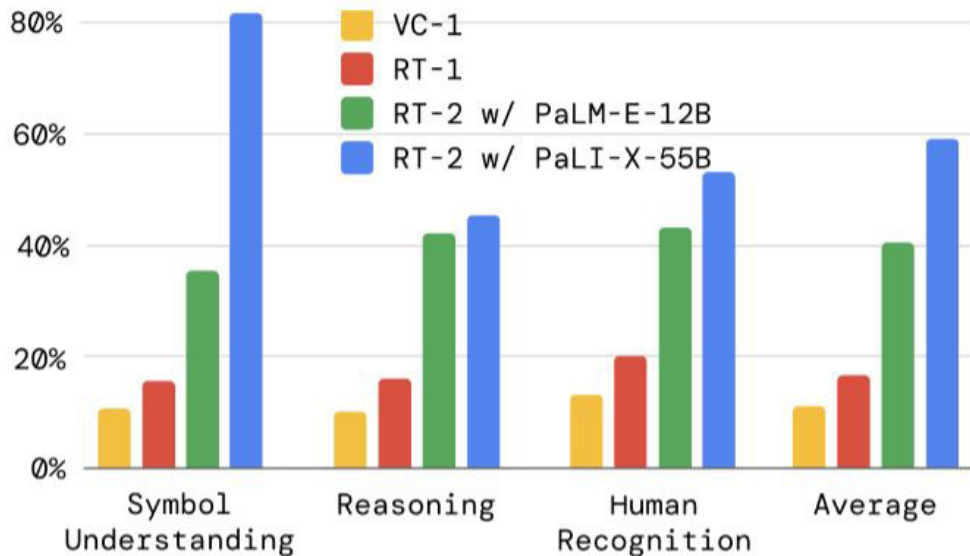
How does RT-2 generalize over objects, backgrounds and environments?

Leveraging web data enables robustness



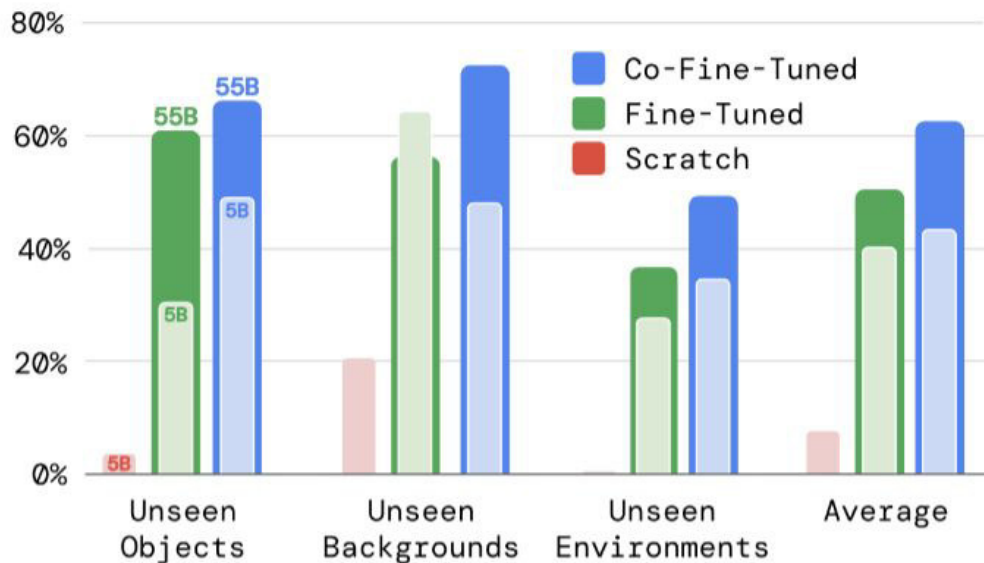
Emergent Skills

- “Put the *strawberry* in the correct *bowl*”
- “Pick up the *bag about to fall off the table*”



Ablations

- Co-fine-tuning is important; 5B performance somewhat comparable with 55B



RT-2 with CoT

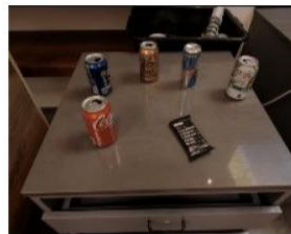
Prompt:
Given Instruction:
Bring me a drink.
Prediction:
Plan: pick 7up can.
Action: 1 143 129 123 145
114 115 127



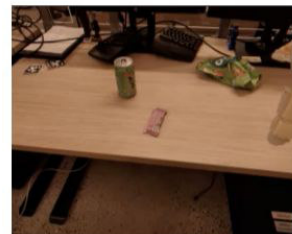
Prompt:
Given Instruction:
Move all the objects
together.
Prediction:
Plan: move green can near
green rice chip bag.
Action: 1 128 126 127 135
123 119 127



Prompt:
Given Instruction:
Pick the object that is
different from all other
objects
Prediction:
Plan: pick rxbar
chocolate. Action: 1 128
129 125 131 125 128 127



Prompt:
Given Instruction:
Move the green objects
together.
Prediction:
Plan: move green can near
green rice chip bag.
Action: 1 130 129 121 131
127 128 127



Prompt:
Given I need to
hammer a nail, what
object from the scene
might be useful?
Prediction:
Rocks. Action: 1 129 138
122 132 135 106 127



Figure 7: Rollouts of RT-2 with chain-of-thought reasoning, where RT-2 generates both a plan and an action.

To Summarize

- Continuing the trend: more data -> more performance
- Web data -> emergent skills
 - Human Recognition
 - CoT Reasoning

OpenVLA: An Open-Source Vision-Language-Action Model

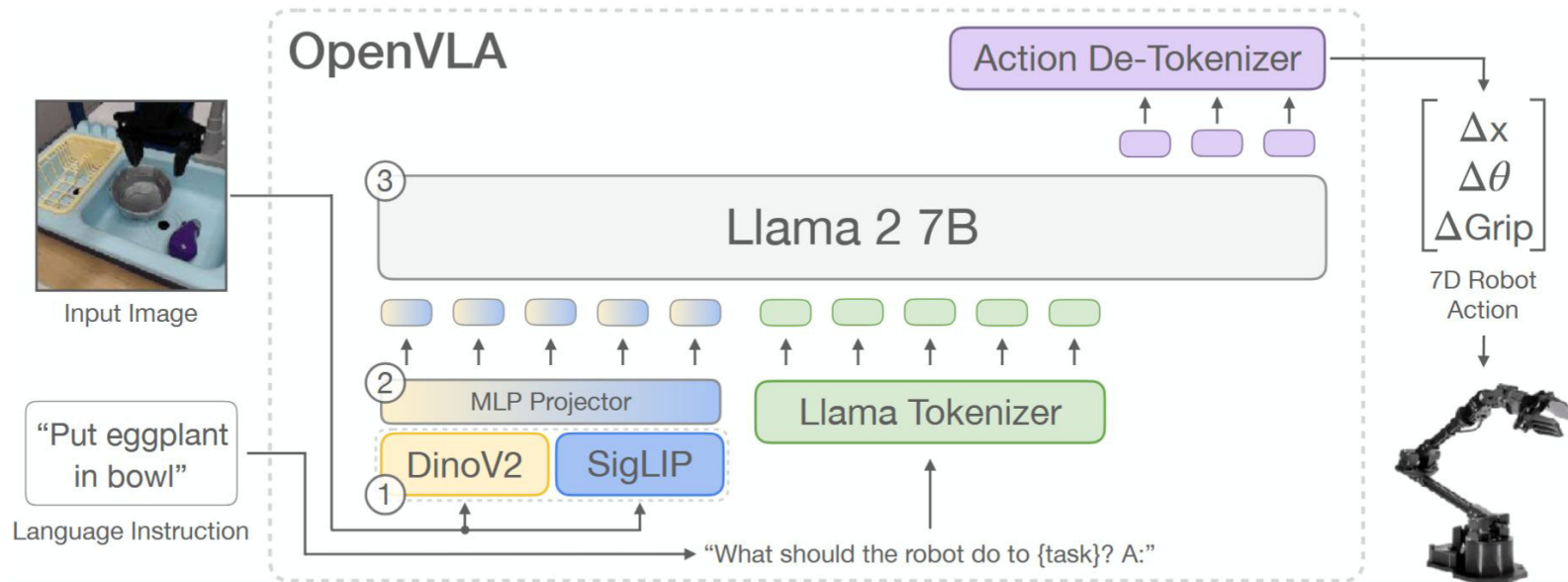
Moo Jin Kim^{*,1} Karl Pertsch^{*,1,2} Siddharth Karamcheti^{*,1,3}
Ted Xiao⁴ Ashwin Balakrishna³ Suraj Nair³ Rafael Rafailov¹ Ethan Foster¹ Grace Lam
Pannag Sanketi⁴ Quan Vuong^{5,†} Thomas Kollar³ Benjamin Burchfiel³ Russ Tedrake^{3,6} Dorsa Sadigh¹
Sergey Levine² Percy Liang¹ Chelsea Finn¹

<https://openvla.github.io>

RT-2 Problems

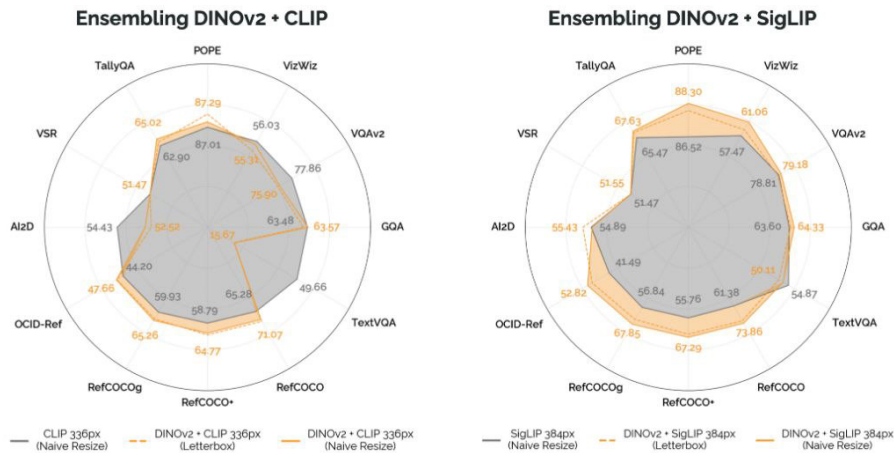
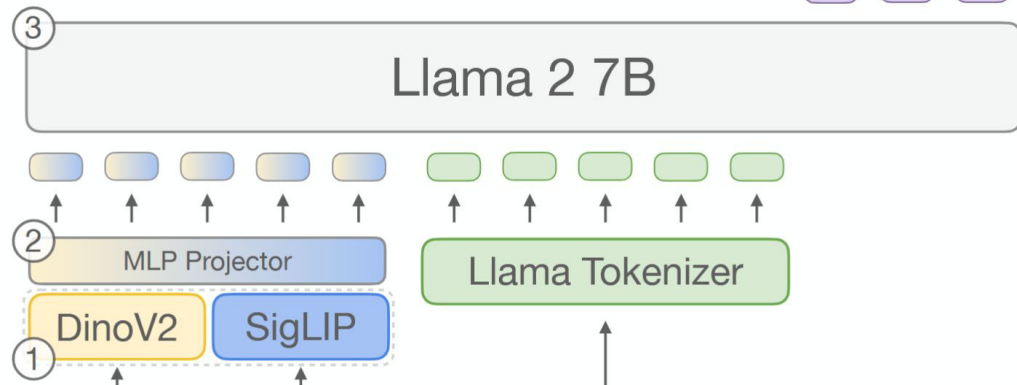
- Absolutely massive
 - RT-2-PaLI-X Parameters: 55B
 - RT-2-PaLM-E Parameters: 562B
- **SoTA is mostly closed-source**
- While robust, there are no best practices for adapting VLAs to new robots, environments, and tasks
- **Let's build a smaller, open-source model that is easy to fine-tune**

OpenVLA



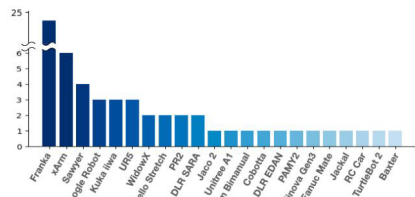
Prismatic VLM

- Uses Prismatic-7B VLM as base
- Concats SigLIP and DINOv2 embeddings
 - Improved spatial reasoning
- VLM fine-tuned on LLaVA 1.5 data mixture

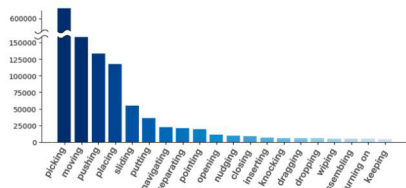


Fine-Tuning

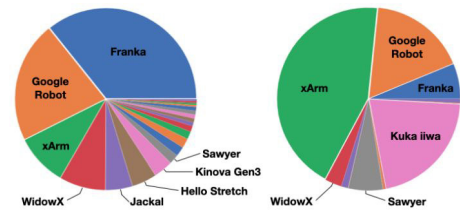
- Procedure is almost exactly same as RT-2
 - Different Tokenizer
 - Not co-fine-tuned
- Difference lies in the data
- Open X-Embodiment Dataset
 - 70 individual robot datasets
 - More than 2M robot trajectories
- Data Processing
 - Only data with at least one 3rd person camera and single arm-effector control
 - Use same data-mixture weights as Octo
-> **emphasize diversity**
 - DROID except for the last 3rd of training
- OpenVLA Codebase
 - Easily deployable fine-tuning code



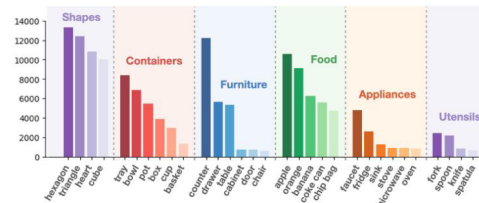
(a) # Datasets per Robot Embodiment



(d) Common Dataset Skills

**(b) # Scenes per Embodiment**

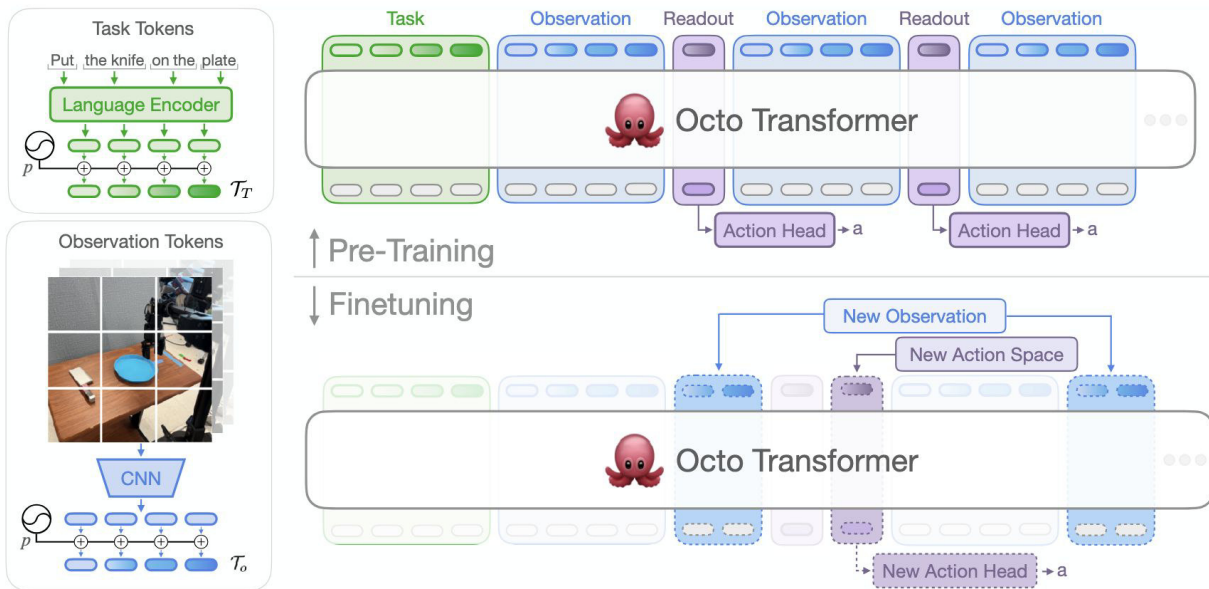
(c) # Trajectories per Embodiment



(e) Common Dataset Objects

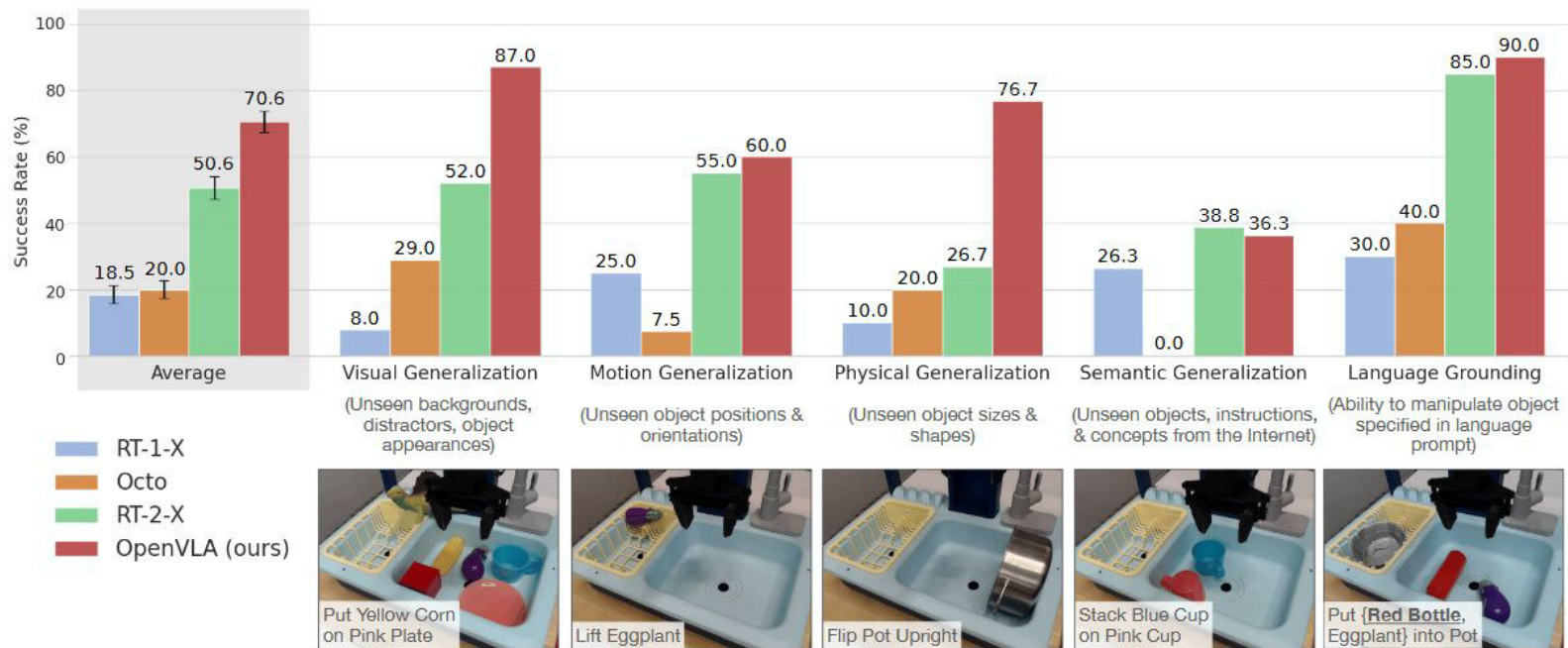
Baselines

- RT-1-X & RT-2-X: Both RT models further finetuned on OpenX
- Octo: Another open-source model

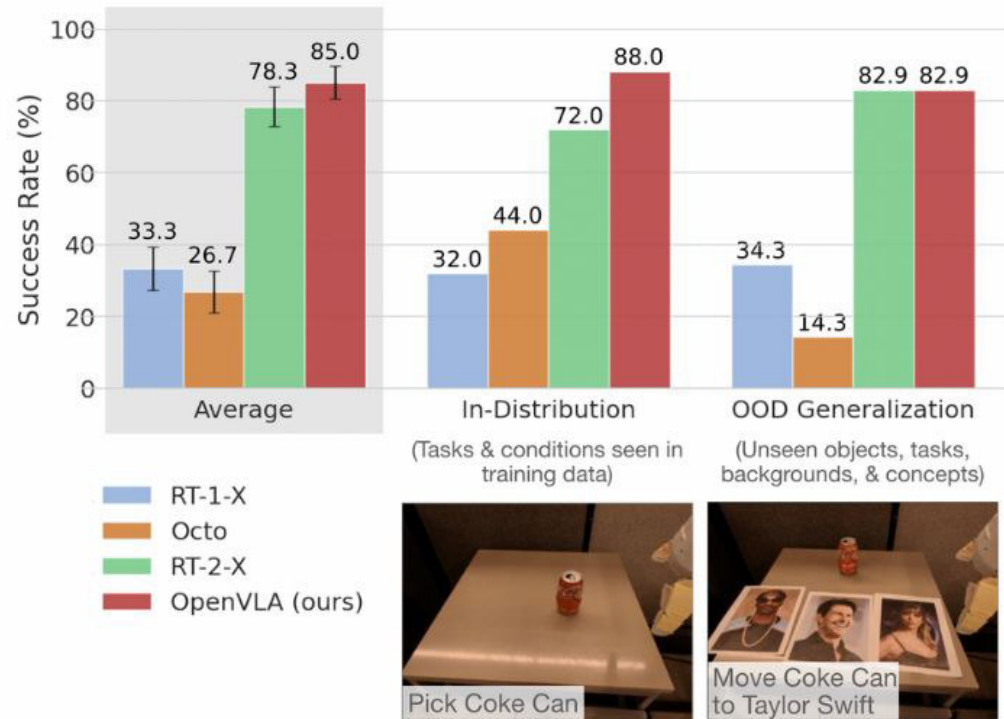


Does OpenVLA generalize?

- RT-2-X trained on more Web data -> higher semantic generalization

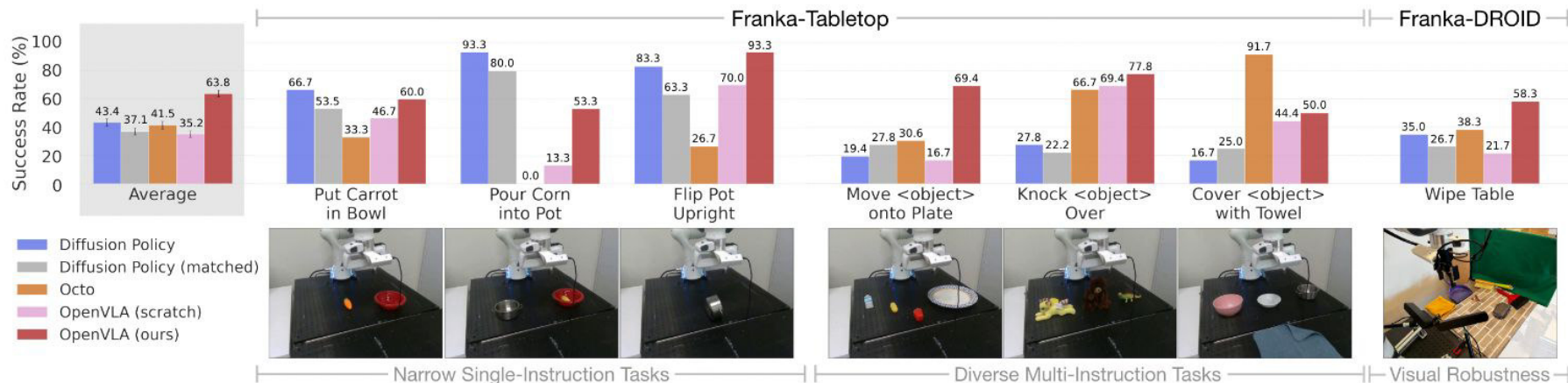


- More data + better cleaned data -> higher generalizability
- Fused vision encoder
 - Semantic + spatial



Adapting to New Tasks & Robot Setups

- OpenVLA scratch - fine-tuned directly on VLM w/o OpenX pretraining
- Diffusion Policy - Data-efficient imitation learning (specialist)
- Specialist better at simple, single instruction tasks; Generalist better in more diverse tasks



Parameter Efficient Fine-Tuning

- Fine-tuning on LoRA has minimal impact on performance

Strategy	Success Rate	Train Params ($\times 10^6$)	VRAM (batch 16)
Full FT	69.7 ± 7.2 %	7,188.1	163.3 GB*
Last layer only	30.3 ± 6.1 %	465.1	51.4 GB
Frozen vision	47.0 ± 6.9 %	6,760.4	156.2 GB*
Sandwich	62.1 ± 7.9 %	914.2	64.0 GB
LoRA, rank=32	68.2 ± 7.5 %	97.6	59.7 GB
rank=64	68.2 ± 7.8 %	195.2	60.5 GB

Effects of Scaling

- More data and more diverse data leads to scaling in performance
- (**Back to DT**) modeling trajectory sequences -> more scale
 - Transformers enable scale
- Leveraging web data overcomes some of the limitations of imitation learning by enabling emergent abilities